



QDCNN-DMN: A hybrid deep learning approach for brain tumor classification using MRI images

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ABSTRACT

Brain Tumor (BT) is the deadliest tumors found globally, which commonly develop in both adults and children. Due to the variety of tumor cells is difficult to categorize a large variety of tumor types, which also leads to more complexity. The key intention of this work is to build and create a useful approach for classifying BT using a hybrid Quantum Dilated Convolutional Neural Network-Deep Maxout Network (QDCNN-DMN). Here, the Magnetic Resonance Imaging (MRI) is used as the input to the image enhancement framework, where it is sharpened using logarithmic transformations. Deep Embedded Clustering (DEC) is used in the skull-stripping process. After that, tumor segmentation is done by the Structure Correcting Adversarial Network (SCAN), and then, the statistical and some other important features are extracted. Subsequently, BT detection and classification are done by the proposed hybrid QDCNN-DMN. The QDCNN-DMN is framed by the incorporation of Quantum Dilated Convolutional Neural Network (QDCNN) and Deep Maxout Network (DMN). It is demonstrated that QDCNN-DMN observed superior specificity, sensitivity, and accuracy with 93.7%, 94.5% and 94.0% based on the Figshare dataset for the BT classification.

1. Introduction

The brain cells grow out of control and against nature in BT. It represents one of the most lethal cancers and life-threatening [1]. Due to its rising prevalence and high death rate across all age groups, BT is one of the worst diseases in the world. There are currently about 120 different forms of tumors, and because the brain has such a complicated structure. It makes it more challenging to detect them because they all have different shapes and sizes when they first occur. For many years, several medical imaging modalities consisting of CT scans, PET scans, MEG, and MRI have been used to identify brain defects. MRI multimodality imaging technology is the most widely used and successful method for routinely identifying BT because of its capacity to differentiate between tissue and structure based on contrast levels. Currently, the majority of abnormality detection by MRI is manual, and doctors must spend a lot of effort locating and segmenting tumors for surgical and therapeutic. This manual technique might endanger life and is also susceptible to errors [2]. Due to the tumor's changing appearance, location, shape, and size, the early discovery of BT assists radiologists in a successful prognosis [3]. MRI is utilized specifically to find tumors and track their growth. In

the course of detecting brain malignancies, radiologists analyse various MRI slices.

Tumors that are found early on can be treated more effectively, which improves many patients' chances of living. Nevertheless, radiologists frequently deal with a large volume of MRI data as well as numerous complicated tumors. The result is a lengthy treatment process with a considerable potential for error, particularly when only small slices are impacted [4]. Additional data from MRI, such as relaxation durations, flow velocity, and chemical change, would contribute to a more accurate description of brain tissue. By using fully automatic recognition and analysis, it is possible to increase the ability of doctors to diagnose and decrease the time needed for an accurate analysis [5,6,7]. For the clinical diagnosis of BTs, MRI is becoming more and more popular. Currently, tumor analysis and segmentation rely on images obtained from different MRI sequences [8,9,10].

To increase the quality of images, image processing and image enhancement approaches are applied in the medical industry [11,12,13,14]. Threshold methods and contrast adjustment are used to attract focus to certain features in MRI images. For the classification and recognition of BT, the procedures of edge detection, histogram,

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Table 1
Related works.

References	Models	Merits	Demerits
Ghanavati <i>et al.</i> [25]	Adaboost classifier	1. Increases the method's resilience. 2. High accuracy.	1. High processing time.
Mallick <i>et al.</i> [26]	Autoencoder	1. Training error is less. 2. Faster training times	1. Low accuracy.
Kumar and Mankame [27]	Deep CNN	1. Less complexity. 2. High classification accuracy.	1. High processing time.
Raza <i>et al.</i> [28]	Hybrid DeepTumorNet	1. Better accuracy.	1. Time complexity is high. 2. Low classification performance.
Ait Amou <i>et al.</i> [4]	Bayesian optimization-based CNN	1. Training performance is high.	1. Limited number of images.
Amran <i>et al.</i> [29]	Hybrid Deep Tumor Network	1. Less training time.	1. Efficiency is low.
Shanthi <i>et al.</i> [5]	OHDNN	1. Better accuracy. 2. Low prediction error.	1. Only using the MRI images for tumor detection.
Kurdi <i>et al.</i> [30]	HHOCNN	1. Less complexity. 2. Decreasing the segmentation problem.	1. Low prediction output.
Bhujanga Rao <i>et al.</i> [31]	Modified U-Net structure-based residual networks	1. Overlapping is reduced.	1. High computational complexity.
Sandhiya <i>et al.</i> [32]	Enhanced learning machine	1. Classification accuracy is high.	1. Processing time is high.
Parkhi <i>et al.</i> [33]	CNN	1. High efficiency. 2. Accuracy is high.	1. High healthcare cost.

segmentation, and morphology are essential [15]. A popular technique that has been utilized recently to create an automatic, semiautomatic, or hybrid model to accurately classify [16] and segment tumors in less time is DL which is a branch of ML [3]. The DL methods are efficaciously used for high-dimensional databases. To address the aforementioned issues with multiclass brain tumors classifications, such as the similarity between tumor types, the removal of crucial information, and high-dimensional datasets, automated techniques utilizing DL are established [17,18,19,20]. The most recent multimodal BraTS tests also showed that DNNs performed very well. Therefore, efficient therapy of BT depends on precise diagnosis and categorization. The transfer learning technique is generally used to save time when there is a shortage of data and processing resources. This method uses the leveraging knowledge gained from one task to improve performance on related tasks [21,22,23,24].

The major contributions of this work are:

- This research developed the QDCNN-DMN model for the classification of BT.
- The Logarithmic transformation technique is used for image enhancement. After that, the Skull stripping process is done by using DEC. The segmentation process is done by using the SCAN.
- Finally, extracted the features to detect the tumor by utilizing the QDCNN-DMN, which is a combination of QDCNN and DMN.

The rest sections in this work are ordered as follows; the inspiration for developing this model, conventional BT detection, and classification approaches are designated in section 2. The section 3 explains the QDCNN-DMN. Moreover, section 4 illustrates the results and discussions. Lastly, section 5 provided the conclusion and future scope of this model.

2. Literature and motivation

Ghanavati *et al.* [25] devised an Adaboost classifier for BT detection. This technique achieved a significant probability of tumors. Additional images of patients were not added to increase the method's resilience and accuracy. Mallick *et al.* [26] introduced an Autoencoder for Brain MRI image classification. This approach allowed for less training error, faster training times, and less effort. Nevertheless, it did not investigate the potential of fusing the DNN with numerous different autoencoder variants, so the accuracy of this model was decreased. Kumar and Mankame [27] established Deep CNN for BT classification. This method decreased the complexity of processing and improved classification accuracy. Nevertheless, it failed to notice the Statistical and LBP features along with the HoG features. Raza *et al.* [28] established a Hybrid DeepTumorNet model for the classification of BT. This technique calculated precise features for brain classification as well as extra descriptive and discriminative information. The effectiveness of the hybrid technique was not considered in the other types of analysis of medical images.

Ait Amou *et al.* [4] developed Bayesian optimization-based CNN for BT classification. Here, the optimized model showed that training performed smoothly and there was no overfitting problem. This technique failed to be considered in real-world clinical applications, which was considered as major challenge. Amran *et al.* [29] introduced a Hybrid Deep Tumor Network for BT classification and detection. In this case, the training period for the suggested model was decreased, which improved the effectiveness of the learning process. However, it failed to evaluate the efficiency of this technique with various categories of biomedical imaging. Shanthi *et al.* [5] devised an OHDNN for automatic BT classification. Here, this approach attained superior accuracy and minimized the variance of prediction error. However, this approach failed to concentrate on the map reducer architecture. Kurdi *et al.* [30] introduced HHOCNN for BT classification. This method lowered the complexity of identifying edge details and minimized uncertainty problems while segmenting the region. However, it did not include an improved feature selection and learning model to enhance BT analysis.

Bhujanga Rao *et al.* [31] introduced the modified U-Net structure based on residual networks for the detection of BT. The primary benefit of this approach was the prevention of de-convolution overlap. However, it also introduced a significant computational complexity. Sandhiya and SKS Raja [32] developed an enhanced learning machine for the classification of brain tumors. This approach used the pre-trained Inception V3 and DenseNet201 models to extract fundamental features. The primary benefit was enhancing classification accuracy. However, it also incurred a significant processing time. Parkhi *et al.* [33] introduced the CNN model for the detection of brain tumors. In this approach, abnormal images and the segmentation of tumor regions were identified using multi-level thresholding. This approach was used to improve accuracy and efficiency. In this method, the cost was high.

Table 1 explains the related works of the traditional approaches.

2.1. Motivation

A brain tumor refers to an abnormal mass of rapidly growing cells within the brain or skull. It can lead to several symptoms, including headaches, mood fluctuations, bladder control issues, and memory impairment. MRI-based brain tumor classification has become a crucial area in medical imaging due to its non-invasive and highly detailed visualization capabilities, particularly for soft tissues like the brain. MRI provides high-resolution images, especially valuable for observing the structural details of soft tissues, enabling detailed assessment of tumors' size, shape, and location. However, this field faces several characteristics and challenges that affect the accuracy and reliability of classification models:

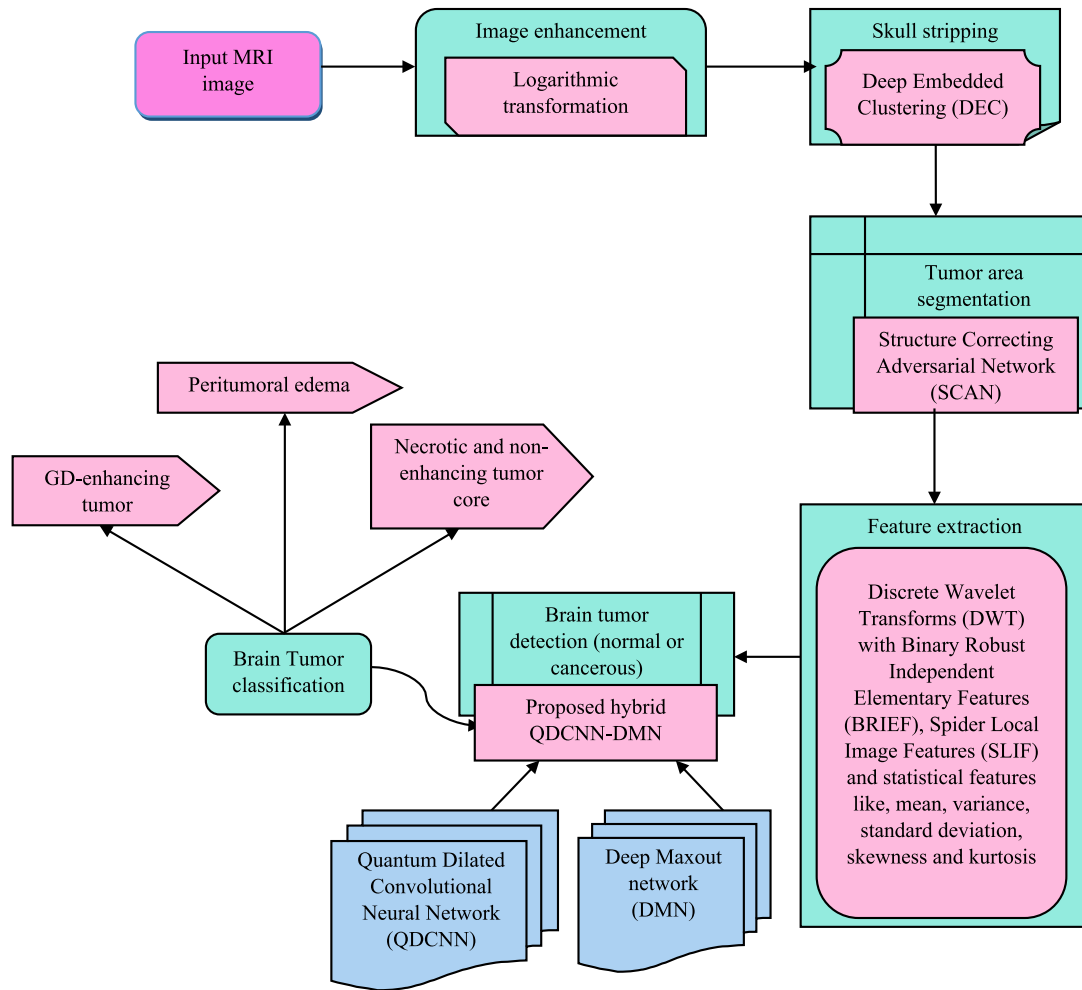


Fig. 1. Block diagram of proposed QDCNN-DMN for brain tumor classification.

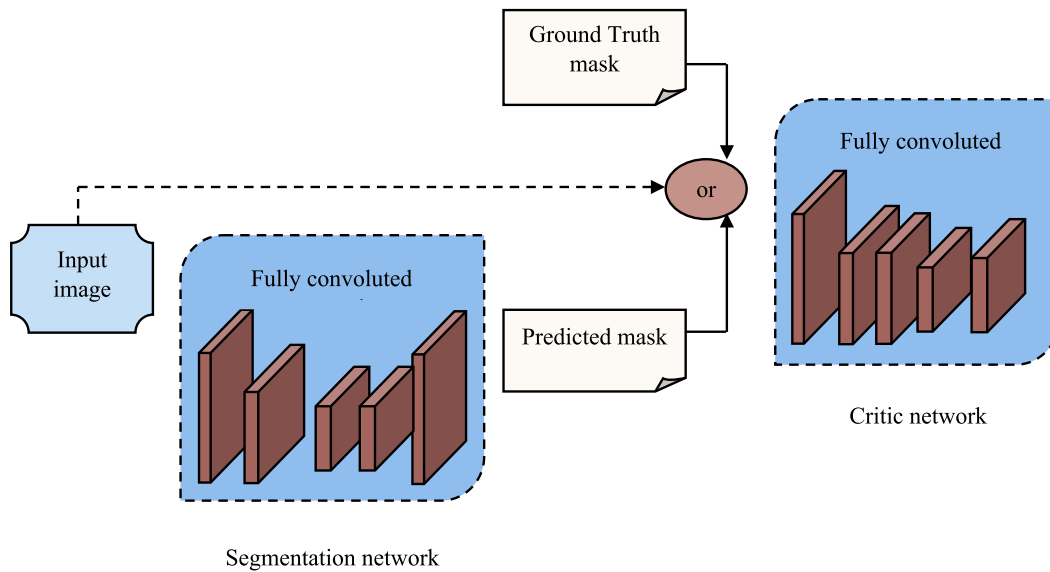


Fig. 2. Overview of the SCAN framework.

- Brain tumors differ widely in form, location, size, and type. This variability complicates the classification process.
- Accurate segmentation of tumor regions is challenging due to the presence of noise and similar intensity values of adjacent healthy tissues.

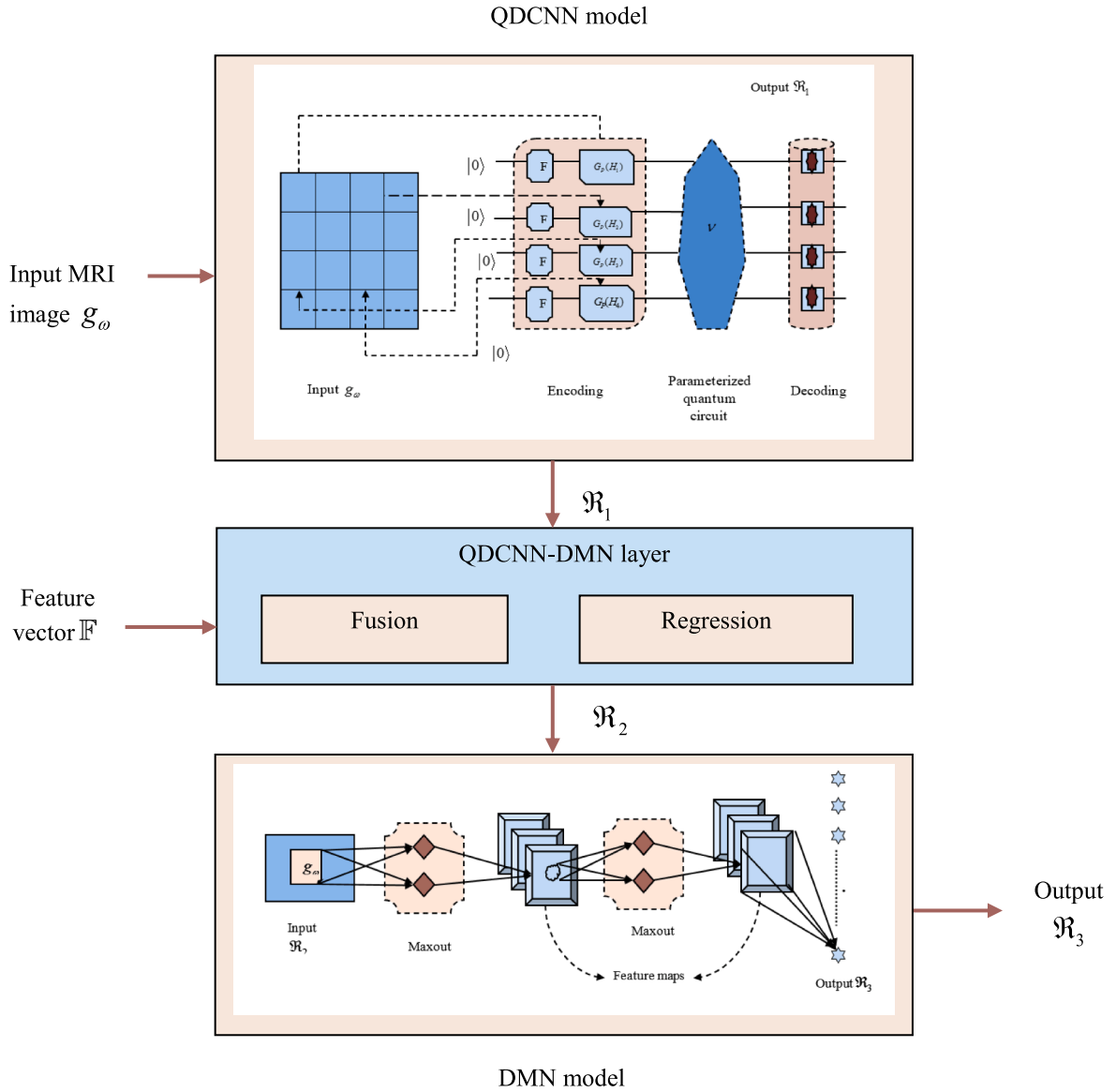


Fig. 3. Structural diagram of the proposed QDCNN-DMN model.

- MRI datasets for brain tumors are often limited, which leads to imbalanced training data. This can result in overfitting and reduced generalizability of classification models.
- MRI images include both relevant and irrelevant features for classification, so feature selection and extraction methods are critical to minimize noise and optimize model performance.
- MRI data is high-dimensional, and processing large volumes of detailed image data is computationally intensive.

These issues serve as the motivation for developing the QDCNN-DMN for BT classification. In this research, the QDCNN has dilation properties for allowing to capture of multi-scale features across varying tumor shapes, sizes, and locations. This flexibility enhances its adaptability to different tumor types, improving classification accuracy despite tumor variability. Also, the SCAN mechanism is used for tumor segmentation, which effectively distinguishes tumor regions from adjacent tissues with similar intensities. This approach mitigates noise and improves the precision of tumor boundaries, leading to more reliable segmentation. Moreover, the DEC helps to optimize feature extraction even with a limited dataset, which enhances the generalizability of the model and reduces overfitting issues. Additionally, the QDCNN reduces the

computational complexity by skipping certain convolutions without losing important spatial information. The QDCNN-DMN minimizes the data processing load, allowing the model to handle high-dimensional MRI data effectively while maintaining performance. Thus, the proposed method overcomes the limitations of the existing schemes.

3. Proposed QDCNN-DMN for BT classification

One of the major medical conditions connected with abnormalities in the human brain is BT. It is a cluster or mixture of unnatural brain tissue. Fig. 1 exhibits a schematic representation of QDCNN-DMN for BT classification using MRI images. This work develops the QDCNN-DMN to detect the BT on MRI images. Initially MRI image taken as input from the dataset is given to the image enhancement module where the image is sharpened using Logarithmic transformations [34]. Then, the process of Skull stripping is done using DEC [35]. After that, the tumor area is segmented using SCAN [36], and then, the process of feature extraction is carried out to extract some important features such as DWT [37] with BRIEF [38], SLIF [39] and statistical features like, mean, variance, standard deviation, skewness and kurtosis. Finally, tumor detection is done by using the QDCNN-DMN, which is the combination of QDCNN

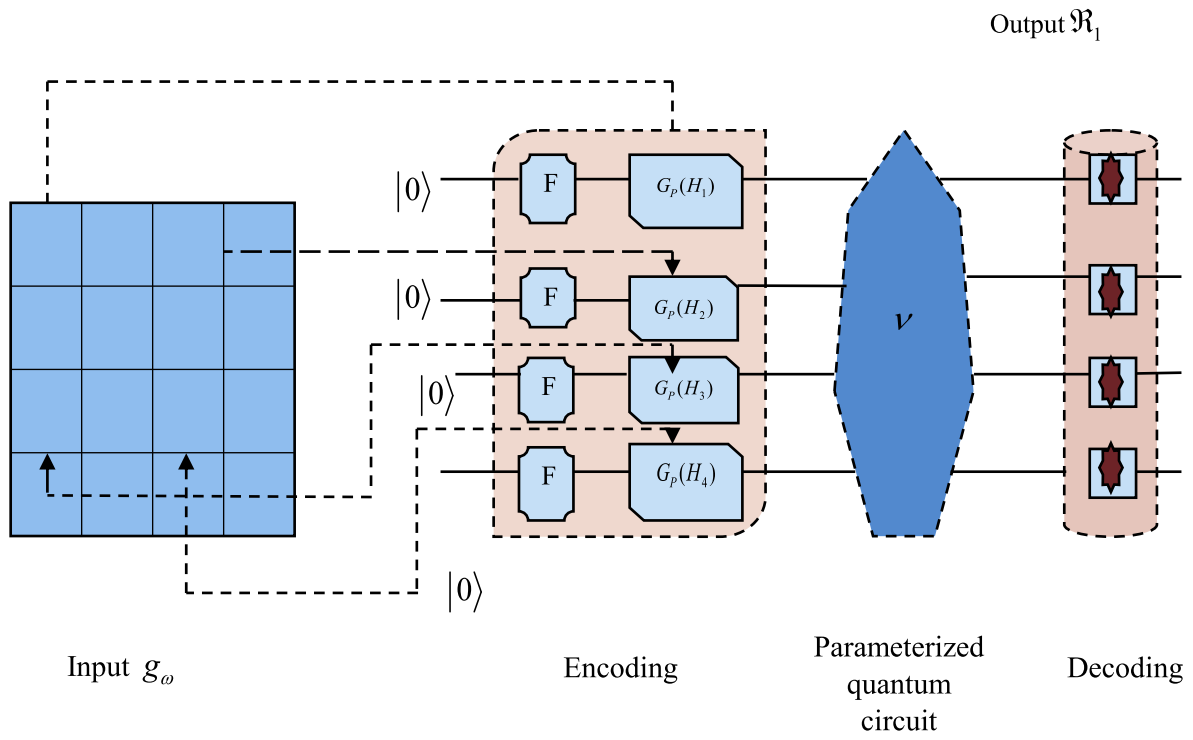


Fig. 4. The architecture of the QDCNN framework.

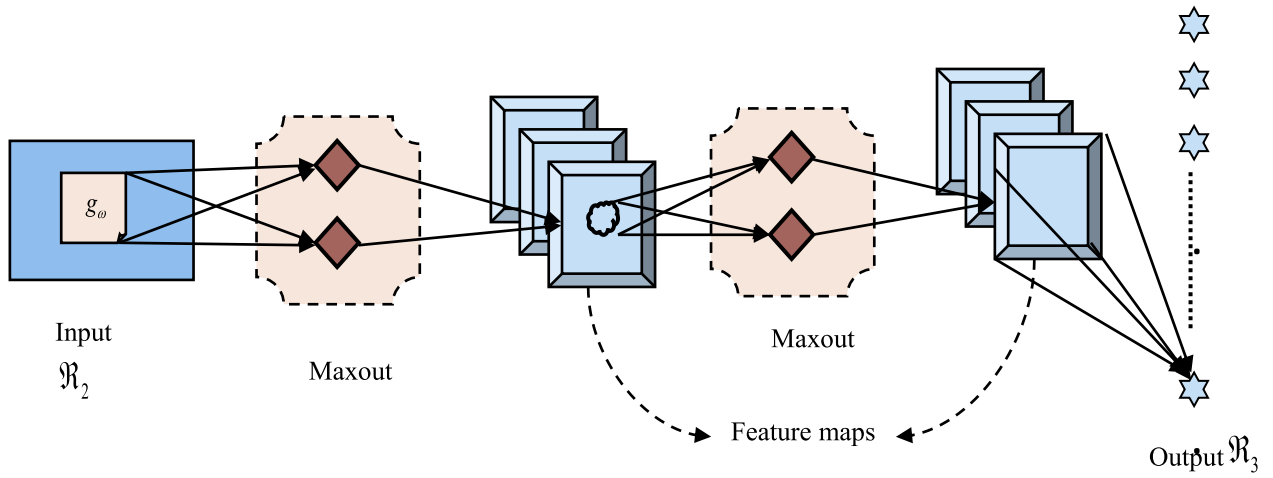


Fig. 5. Architecture of DMN model.

Table 2
Parameter Details.

Parameters	Values
QDCNN	
Loss	Categorical cross-entropy
epochs	20
Batch size	32
Learning rate	0.001
DMN	
Epochs	20
filters	50
Loss	Categorical cross-entropy
Kernel size	3
Batch size	32

[40] and DMN [41] where layers are modified and the tumor is further classified into three types like GD-ET (level-1), ED (level-2), and NCR/NET (level-3) using the QDCNN-DMN.

3.1. Image acquisition

Consider an MRI input image for the classification of BT that was obtained from the specific dataset stated in [42,43]. It can be demonstrated by,

$$N = \{g_1, g_2, \dots, g_\omega \dots g_N\} \quad (1)$$

From the above equation, overall images in the database N are signified, while g_ω representing ω^{th} input images.

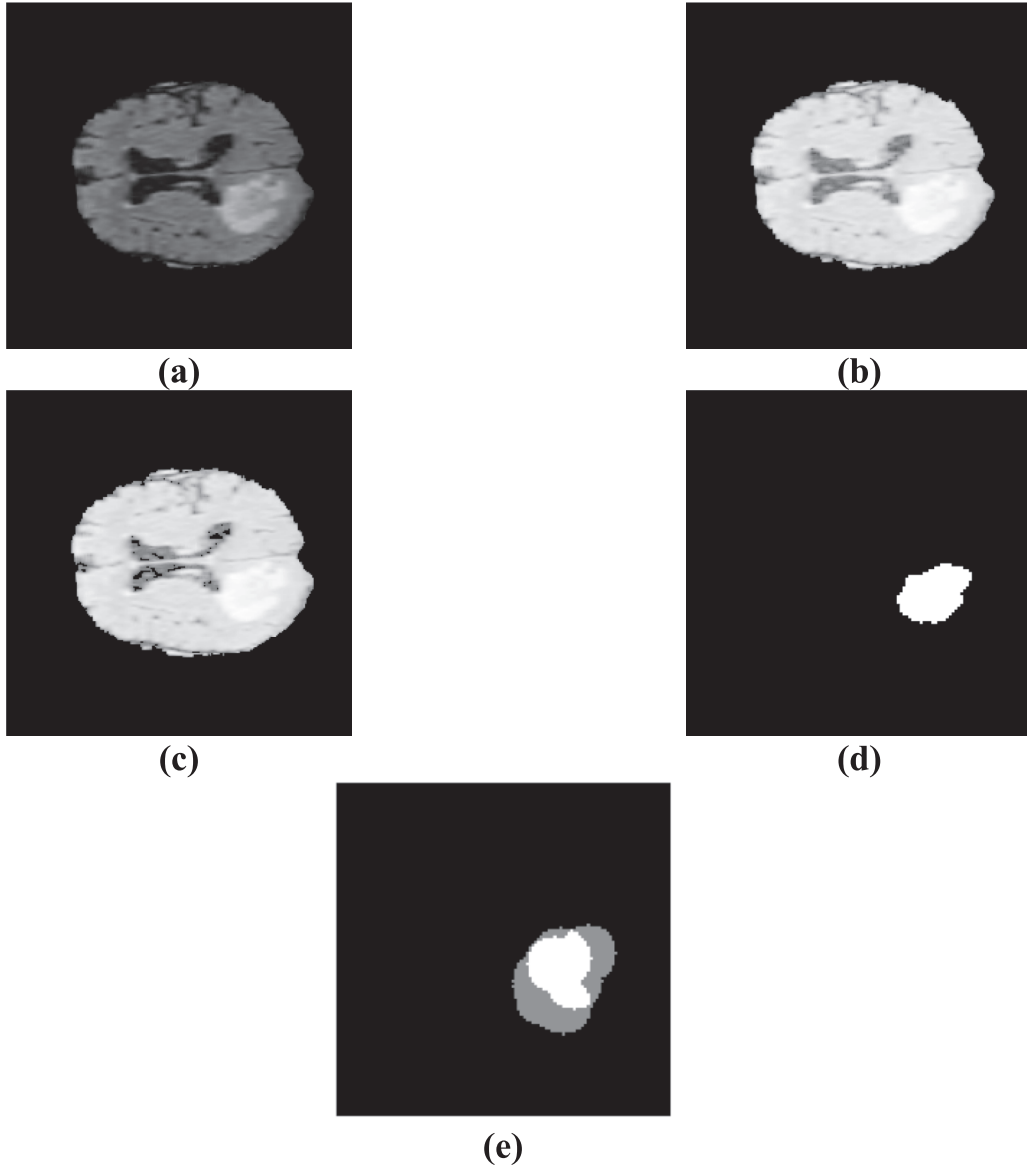


Fig. 6. Experimental results of BT using BRATS 2018 dataset, a) input, b) enhanced image, c) skull stripping image, d) segmented image and e) SLIF feature image.

3.2. Image enhancement

The image input g_o is used to enhance the image. The main aim of image enhancement is to alter the features of images to suit a task and the specific observer. Logarithmic transformations play a crucial role in enhancing images by adjusting pixel values. Log transformation expands the representation of dark regions in the image. By emphasizing low-intensity pixels, it enhances contrast and reveals subtle details. It adjusts the overall brightness while preserving relative differences. Logarithmic transformations enhance the validity of statistical tests, stabilize variance, and yield more reliable regression results. Additionally, they simplify complex relationships, facilitating easier interpretation and modelling. To improve the input image, logarithmic transformation [34] is applied here.

Following is a general representation of the log transformation.

$$\zeta = \lambda * \log(1 + M) \quad (2)$$

Where, pixel values are signified as ζ and M , the number of columns is symbolized as λ . Wider ranges of output values are produced from a narrow range of low input grey-level values using the log transformation. The opposite transformation is done utilizing the inverse log

transformation. When there may be a very wide range of values in the input grey-level values, log functions are especially helpful. The output attained from the logarithmic transformation is denoted as g_e .

3.3. Skull stripping using Deep embedded clustering

An essential technique for both data analysis and visualization is clustering. DEC [35] is frequently used due to its enhanced scalability and performance. The DEC achieves simultaneous feature representation and cluster assignments. Optimizing these objectives significantly improves results compared to separate optimization steps. Additionally, DEC leverages the capabilities of deep learning to enhance clustering performance. In this case, the enhanced image g_e is contemplated as the input to DEC, to perform partitioning to transform the data, a non-linear mapping is employed. A low-dimensional feature space is created by mapping the input data to maximize the clustering objective. Here, two steps are involved: a) deep autoencoder is employed in parameter initialization, and b) in parameter optimization, where an iterative approach is utilized to reduce KL divergence and compute an auxiliary target distribution. Consider that there are an overall of i points $\{R_c \in \beta\}_{c=1}^R$ that have to endure partition into x clusters, which are

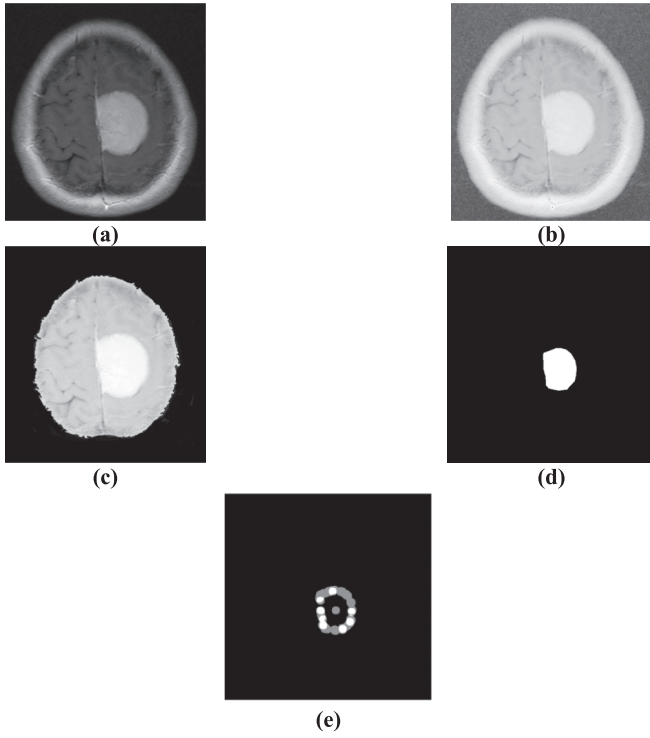


Fig. 7. An experimental outcome for BT utilizing Figshare dataset, a) input MRI image, b) enhanced image, c) skull stripping image, d) segmented image and e) SLIF feature image.

represented by a centroid ξ_δ , $\delta = 1, \dots, x$. DEC employs a non-linear mapping $D_o : \beta \rightarrow B$ to convert the data hidden feature space B , and o indicates learnable parameters. DNN is used for parameterizing D_o . The DEC synchronously learns DNN parameters o and x cluster sets in feature space B .

3.3.1. Clustering with KL divergence

To improve the clustering provided a primary value of o and $\{\xi_\delta\}_{\delta=1}^x$, an unsupervised technique that alternates between two phases is used. In the initial stage, a soft assignment among the embedded point and cluster centroid is established. Subsequently, the deep mapping D_o is reorganized and current high-confidence distributions for decontaminating centroids of clusters are learned using the distribution of auxiliary targets. At last, the convergence is achieved, otherwise, the process is repeated.

a) Soft assignment

The Student's t -distribution serves as a kernel to measure the proximity between an embedded point b_c and its corresponding centroid ξ_δ as shown below,

$$Y_{c\delta} = \frac{\left(1 + \|b_c - \xi_\delta\|^2 / \wp\right)^{-\frac{\wp+1}{2}}}{\sum_{\delta} \left(1 + \|b_c - \xi_\delta\|^2 / \wp\right)^{-\frac{\wp+1}{2}}} \quad (3)$$

Where in, $Y_{c\delta}$ states soft assignment, which is the probability of allotting a sample c to a cluster δ , degree of freedom indicates $\wp$ and $b_\delta = D_o(R_c) \in E$ is connected with $R_c \in \beta$ after embedding.

b) Reduction of KL divergence

The learned data is used to continuously refine the clusters using a target of auxiliary distribution. To train the model, soft assignments are coordinated to the desired distribution, and KL divergence loss is expressed in the following way.

$$Loss = AB(T||I) = \sum_c \sum_{\delta} e'_{c\delta} \log \frac{e'_{c\delta}}{Y_{c\delta}} \quad (4)$$

In this case, T indicates target distribution, and $Loss$ designates loss of KL divergence between the distribution of the target $e'_{c\delta}$ and soft assignment $Y_{c\delta}$. Moreover, Y_c is initially elevated to the second power, and its normalization is achieved by frequency per cluster to calculate the value e'_c as shown as follows,

$$e'_{c\delta} = \frac{Y_{c\delta}^2 / f_{\delta}}{\sum_{\delta} Y_{c\delta}^2 / D_{\delta}} \quad (5)$$

Here, $D_{\delta} = \sum_c Y_{c\delta}$ denotes soft clusters. As a result, data is efficiently divided into multiple clusters $X_1, X_2, \dots, X_x$, by the DEC. Moreover, the obtained from skull stripping is represented g_{strip} , which is then used in the subsequent step.

3.4. Tumor area segmentation using SCAN network

After skull stripping, the SCAN [36] is utilized for isolating the tumor region. SCAN is a powerful technique for segmenting the BT on MRI images. The SCAN network contains 12 layers of convolutional 2D, 3 layers of max-pooling 2D, and 2 layers of Upsampling 2D. SCAN model has two mechanisms spatial and channel attention mechanisms. The spatial attention mechanism focuses on the relevant spatial regions. Moreover, in the channel attention mechanism, the SCAN adjusts the channel-wise feature responses to enhance the representation of tumor-specific features. Then, the SCAN combines the spatial and channel attention maps. Finally, the result of attention-guided features is used for tumor segmentation. When compared to other methods, the SCAN produced realistic and precise segmentation results. Moreover, the SCAN demonstrates strong adaptability across diverse scenarios, showcasing its robustness and generalization capabilities. The GAN concept is the foundation of SCAN. The SCAN employs the pre-trained FCN to produce a GAN. Additionally, it trains a critic network that receives a ground truth mask, predicted mask, and if desired, original image. The architecture of the critic network is the same, which leads to a similar capacity. The skull stripping output g_{strip} is fed as input to this tumor area segmentation.

Segmentation network: The segmentation network is an FCN, and many of the most advanced semantic segmentation models have this as their fundamental feature [4445]. The outstanding capacity of CNN to extract high-level representations appropriate for dense categorization is partly responsible for the success of FCN. The down- and up-sampling methods are the two modules that are present in the FCN. With architecture similar to that of image classification, the down-sampling process comprises max or average pooling layers and convolutional layers. Usually at a reduced spatial resolution, the downsampling route can mine the high-level semantic information. The path of up-sampling uses the output of the path of down-sampling to predict scores for each class at the pixel level. It also comprises of convolutional and de-convolutional layers and it is also known as transposed convolutional.

Critic network: The critic network is an FCN that is constructed like that of the segmentation network. A similar range of views and comparable model capacity are found in the critic network and segmentation network, which is significant given the enormous object size in the MRI images. An easier and more cost-effective way to add the image to the critic network is that it can optionally accept the original MRI image as an additional channel for input. Including the original MRI image does not boost performance, according to a first experiment. The output gained from the SCAN network is depicted as $\mathbb{S}_T$. An adversarial process is used to jointly train a critic network and a segmentation network in the SCAN architecture, as shown in Fig. 2.

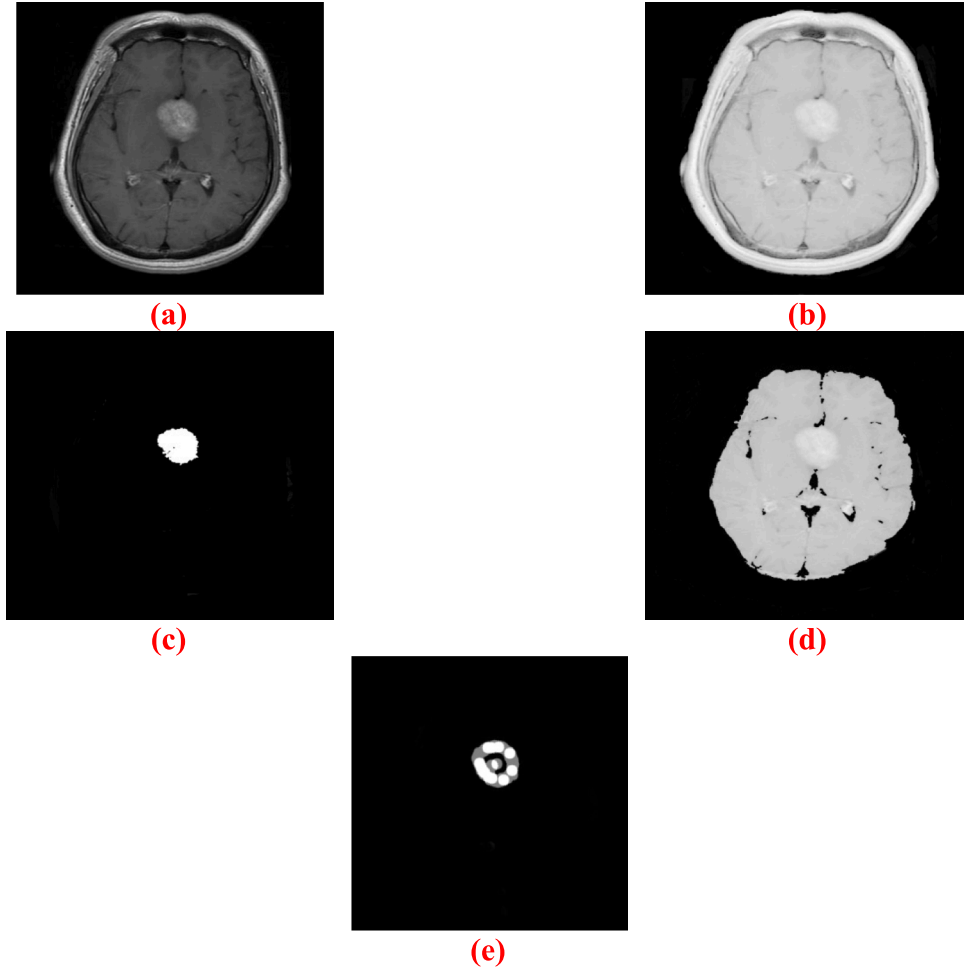


Fig. 8. An experimental outcome for BT utilizing Brain Tumor Classification (MRI) dataset, a) input MRI image, b) Log transform image, c) segment image, d) skull stripping image and e) SLIF feature image.

3.5. Feature extraction

It is used to categorize the patterns since the extraction of features yields related shape information. In image processing, one particular technique for reducing dimensionality is feature extraction. Statistical features, DWT [37] with BRIEF [38], and SLIF [39] are extracted as features from the input MRI image. In this case, segmented output $\mathbb{S}_T$ is specified as an input for mining the features.

a) DWT with BRIEF

The segmented output $\mathbb{S}_T$ is applied to BRIEF, and the obtained output from BRIEF is fed to the DWT feature. The DWT decomposes the input into various levels, like LL, HH, HL, and LH.

i) BRIEF: The original BRIEF extraction method was created for image matching [38]. It randomly chooses a small number of pixel pairs from an image for a binary test and then uses binary coding to summarize the results. These procedures are simple and can be expressed numerically as:

$$s_l = \begin{cases} 1, & \text{if } (\Gamma_k - \Gamma_a) \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

Here, the pair pixel index is denoted as l , and intensity values are indicated as Γ_k and Γ_a .

$$S = \sum_{l=1}^E s_l \cdot 2^{l-1} \quad (7)$$

Where in, the overall number of pair pixels are represented as E .

Despite its ease of use and randomness in the pixel sampling process, BRIEF has been producing excellent results.

ii) DWT: It was applied to brain MR images to derive the coefficient of wavelets. To simplify categorization; the wavelet localizes the frequency information of the signal function [37]. Each of the several frequency components was examined with a resolution appropriate to its scale, and the results were articulated as follows:

$$DWT = \begin{cases} f'_{j,i} = \Sigma s(p) h' * v'(p - 2ji) \\ f'_{j,i} = \Sigma s(p) g' * v'(p - 2ji) \end{cases} \quad (8)$$

The coefficients $f'_{j,i}$ indicate element features in the signal $s(p)$ comparable to the function of the wavelet. The functions $h'(p)$ $g'(p)$ denote coefficients of high-pass and low-pass filters, whereas parameters i j represent factors of translation and wavelet scale. From this LL, HL and HH are combined to create an output $\mathbb{S}_{DBRIEF}$.

b) SLIF

The segmented output $\mathbb{S}_T$ is fed as an input for SLIF. To resolve the image matching issue, SLIF [39] is a new feature descriptor that has been designed. By carefully extracting pixel information from within proximity of each of the previously designated interest points, the SLIF gives a matching set of feature descriptors for the set of interest points. This is gifted by using a unique ORB web sampling technique that was motivated by the simplified orb web mode. With the help of SLIF, low-dimensional feature descriptors that are even more resistant to typical image distortions and transformations can be created. The feature SLIF is denoted as $\mathbb{S}_{SLIF}$.

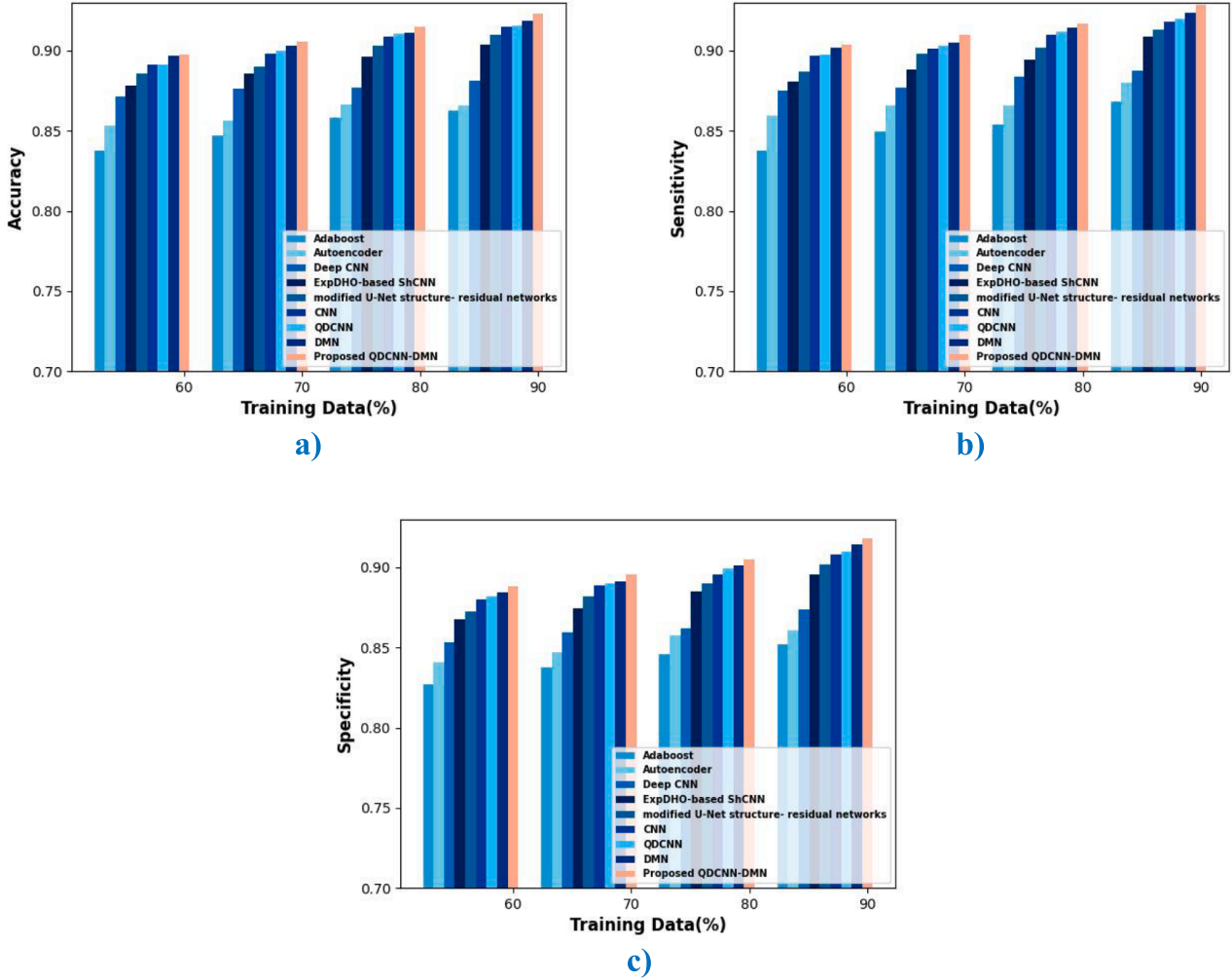


Fig. 9. Comparative evaluation of QDCNN-DMN based on BRATS 2018 using training data, a) specificity, b) sensitivity and c) accuracy.

Furthermore, the output gained from DWT with BRIEF and SLIF is combined and it is represented as $\mathcal{V}_t$.

$$\mathcal{V}_t = \{\mathcal{S}_{DBRIEF}, \mathcal{S}_{SLIF}\} \quad (9)$$

c) Statistical features

Skewness, kurtosis, mean, standard deviation, and variance are some of the statistical features that are mined. In this case, texture feature output $\mathcal{V}_t$ is considered as input to statistical features.

(i) **Mean:** It is the whole values of image pixels partitioned by whole pixel numbers. The formula for calculating the mean is as follows:

$$K_1 = \frac{1}{p} \sum_{Q=1}^p \mathcal{V}_{tQ} \quad (10)$$

Here, p signifies overall images, $\mathcal{V}_{tQ}$ indicates the texture feature output and K_1 designates mean.

(ii) **Variance:** Variance is a feature of how data points differ from the mean. It is articulated as given below.

$$K_2 = V^2 = \frac{1}{p} \sum_{Q=1}^p (\mathcal{V}_{tQ} - K_1) \quad (11)$$

Here, K_2 indicates variance.

(iii) **Standard deviation:** It is the variance in square root, given in the subsequent equation.

$$K_3 = \sqrt{\frac{1}{p} \sum_{Q=1}^p (\mathcal{V}_{tQ} - K_1)^2} \quad (12)$$

where, K_3 represents standard deviation.

(iv) **Kurtosis:** It is used to explain the experimented image distribution concerning the mean.

$$K_4 = \frac{\sum (K_1 - \overline{K_1})^4}{p K_3} \quad (13)$$

Here, K_1 signifies mean, K_3 denotes standard deviation and kurtosis indicated as K_4 .

(v) **Skewness:** The degree to which an image deviates from its original state is its defining characteristic, and it can be shown by,

$$K_5 = \frac{\sum_Q (\mu_p - \overline{K_1})^3}{(p-1) * K_3} \quad (14)$$

Where in, p represents image number while μ_p signifies p^{th} sample and K_5 represented as skewness. Furthermore, statistical features are combined, which is given as follows.

$$K = \{K_1, K_2, K_3, K_4, K_5\} \quad (15)$$

After extracting the texture features and statistical features, it is

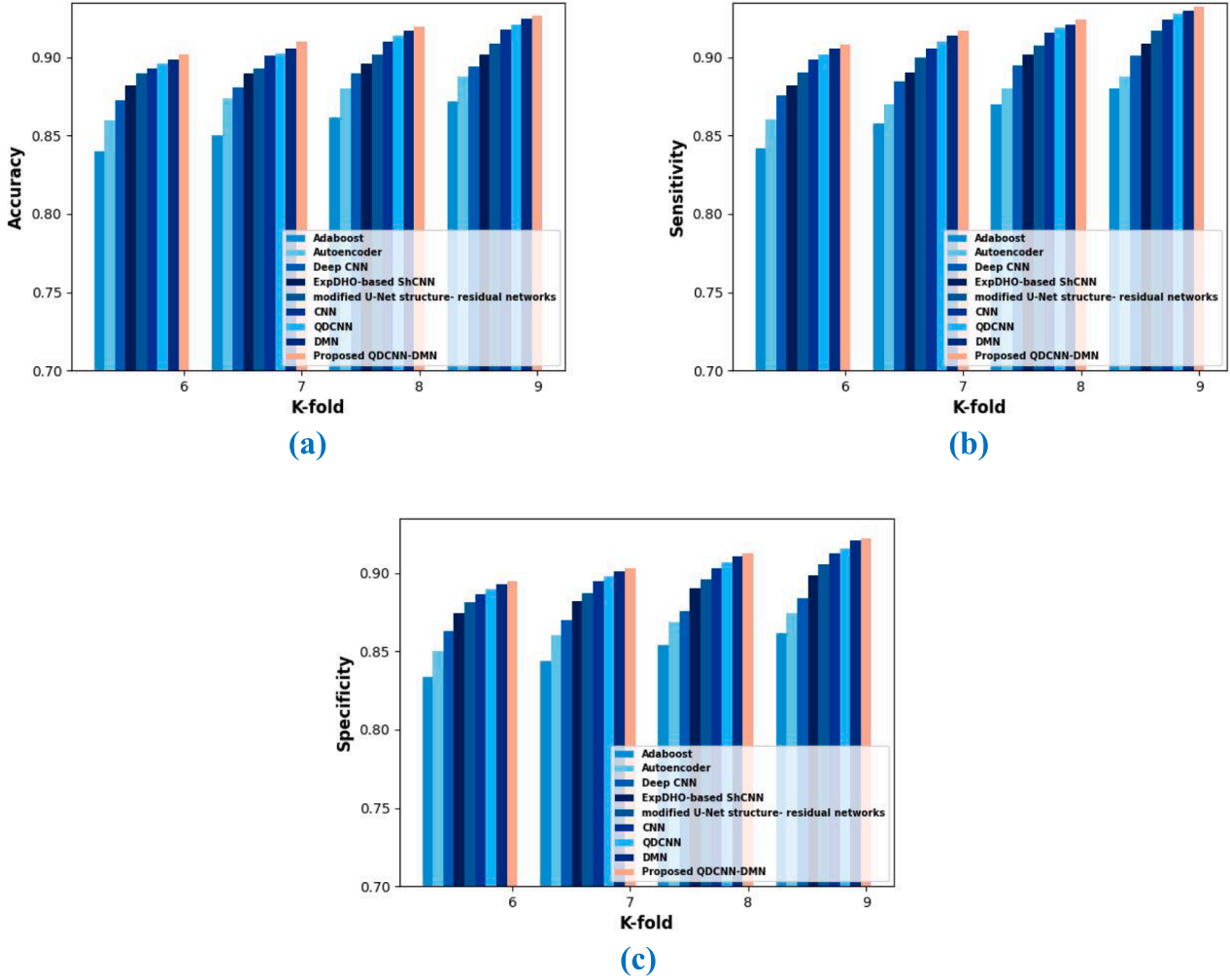


Fig. 10. Comparative evaluation of QDCNN-DMN based on BRATS 2018 using k-fold, a) specificity, b) sensitivity and c) accuracy.

grouped to make a feature vector, which is shown as follows,

$$\mathbb{F} = \{\mathbb{F}_t, K\} \quad (16)$$

3.6. Tumor detection using hybrid deep learning

Anomalous cell formation is the major cause of BT. Early identification of BT can save millions of lives. A higher patient survival rate may result from earlier BT detection with MRI. In this case, QDCNN-DMN effectively detects BT. In addition, QDCNN-DMN is the combination of QDCNN [40] and DMN [41]. An input MRI image g_w is considered for BT detection.

In this work, a hybrid DL approach is developed by combining the QDCNN and DMN based on regression modelling with the help of FC [46]. Initially, the QDCNN model is used to process the MRI image. Then, the fused feature and the obtained output are sent to the QDCNN-DMN layer. The QDCNN-DMN layer is done in both regression and fusion. Estimating the similarity between the targeted and predicted variables is done through regression. At last, the QDCNN-DMN output and fused feature are sent into the DMN model. The expected output is derived from the DMN technique. Fig. 3 shows a general diagram of QDCNN-DMN for BT detection.

3.6.1. QDCNN model

In QDCNNs [40], the QDC layer is quantum. QDCNN is specifically

designed to identify and classify brain tumor images. Its architecture is fine-tuned to enhance performance in this specific domain. The QDCNN framework includes the layers of quantum and classical, and quantum circuit ansatz can be located wherever in this framework. Through the integration of quantum elements, the QDCNN can effectively capture a broader context during the convolution process, leading to enhanced feature extraction. By utilizing QDC, the receptive field can be extended. As a result, kernel size and number of qubits used in QDC are the same as those in ordinary quantum convolution. Throughout the quantum convolution process, QDCNNs are capable of gathering more contexts at a cheaper computational cost.

i) QDCNN composition: The QDCNN framework comprises a single QDC layer and an overall connected layer of 10 neurons. Without any parameters, stride and kernel size for the QDC layer are chosen to be 2×2 and 2 respectively. Especially, a 4-qubit state is encoded from two 2×2 pixels utilizing RY rotation gates. Observe that these 2×2 pixels are not near to one another in the input image because of the QDC. The decoding method functions identically to the encoding approach in that it maps every expected value to a different channel of a single output pixel. Thus, even with a single filter, the quantum layer may create four feature maps from the original 2-D image. This kind of quantum layer permits correlation among the output feature map channels, which could enhance the performance of this framework.

Encoding module: It converts conventional input into a quantum

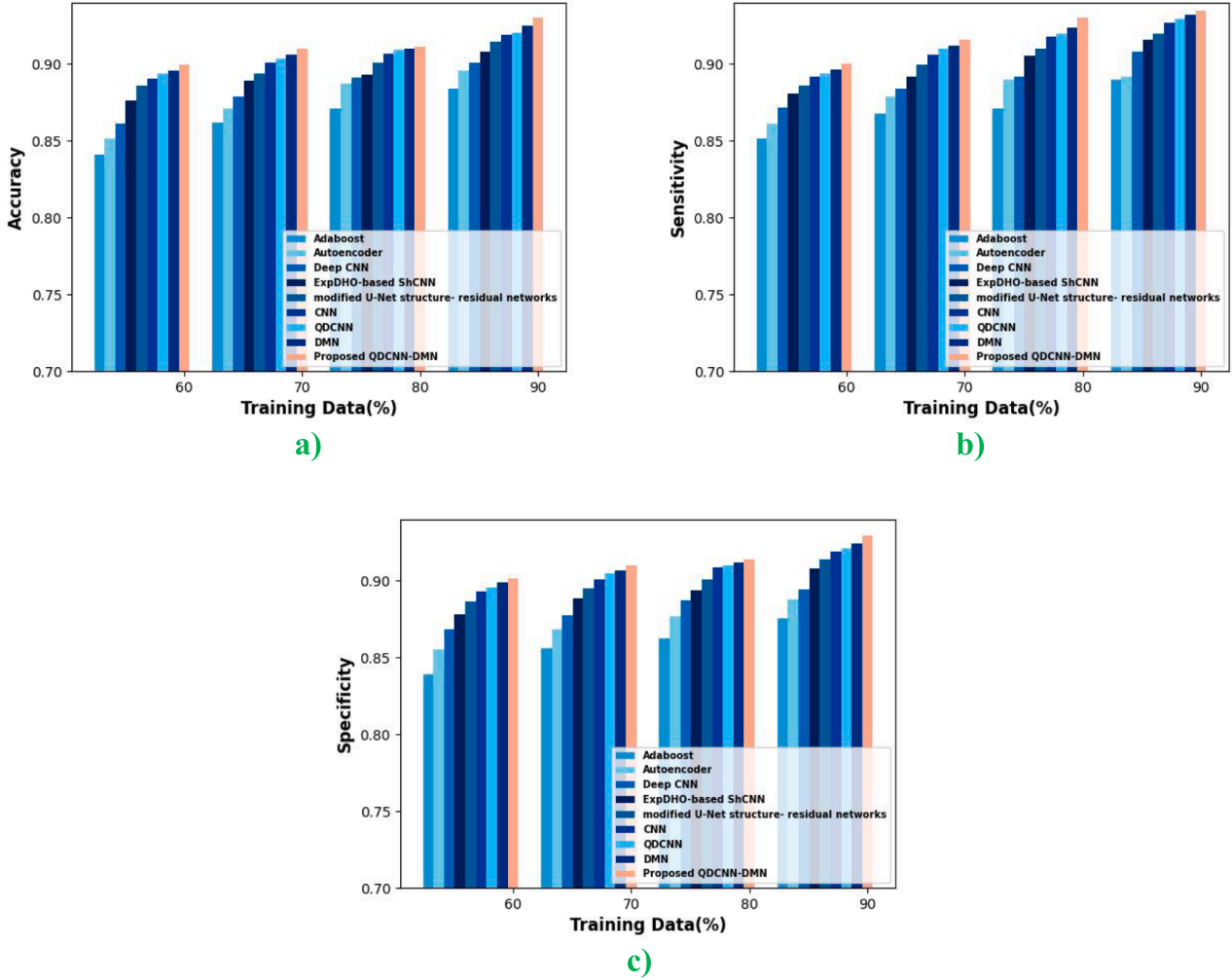


Fig. 11. Comparative evaluation of QDCNN-DMN based on Figshare using training data, a) specificity, b) sensitivity and c) accuracy.

state for subsequent processing in a quantum convolution circuit. The classical input is recognized in this encoding method as the rotational angle of a rotation gate for a single qubit. Let us signify by $m(H)$, the encoding operator which H designates the input vector. The encoded quantum state is then achieved by:

$$|H\rangle = m(H)|0\rangle \quad (17)$$

where, $m(H)$ generally comprises of Hadamard gate which converts the first state into a uniform superposition principle state.

Decoding module: In this module, certain local visible $\Gamma \otimes h$ is measured in the last quantum state $|H, \theta\rangle$ from the entanglement element, where m is same as or less than the overall amount of qubits a in the quantum system. The expectancy-value of the selected noticeable $\Gamma \otimes h$ can be achieved through repetitive dimensions:

$$\mathfrak{R}_2 = k(H, \theta) = \langle H, \theta | \Gamma \otimes h | H, \theta \rangle \quad (18)$$

To mine a classical output vector $k(H, \theta)$, this layer uses a mapping from a quantum state to a classical vector:

$$\varepsilon : |H, \theta\rangle \rightarrow k(H, \theta) \quad (19)$$

Here, H signifies Localization object, θ indicates trainable parameter and $|0\rangle$ denotes vacuum state.

Dilated convolution: It is a category of convolution where the kernel is expanded by generating holes between the next kernel parts.

The dilation rate is defined as the input pixel stride which is sampled in dilated convolution and it is measured by further hyperparameter.

$$j[l, K] = \sum_z \sum_s \ell[z, s] \cdot H[l + z \cdot E, K + s \cdot E] \quad (20)$$

Additionally, for dilated convolution,

$$\kappa_J = \left(\frac{l_J - h - (h - 1)(E - 1) + 2e}{C} \right) + 1 \quad (21)$$

$$\kappa_\eta = \left(\frac{l_\eta - \varphi - (\varphi - 1)(E - 1) + 2e}{C} \right) + 1 \quad (22)$$

Here, E represents dilation rate, κ_J and κ_η denotes spatial resolution, stride is indicated as C and padding is stated as e . The output gained from the QDCNN framework is symbolized as $\mathfrak{R}_1$. Moreover, the structural view of QDCNN is presented in Fig. 4.

3.6.2. QDCNN-DMN layer

After producing output $\mathfrak{R}_1$ utilizing QDCNN, it is employed in the QDCNN-DMN model along with statistical K, DBRIEF $\mathcal{S}_{DBRIEF}$, SLIF $\mathcal{S}_{SLIF}$, and feature vector $\mathbb{F}$. Regression modelling is a statistical method used in this layer to fuse the information based on which features have the greatest influence on the BT detection process. Regression modelling is carried out here by merging the features utilizing the FC [46]. The features are assumed to be obtained at various time intervals. The

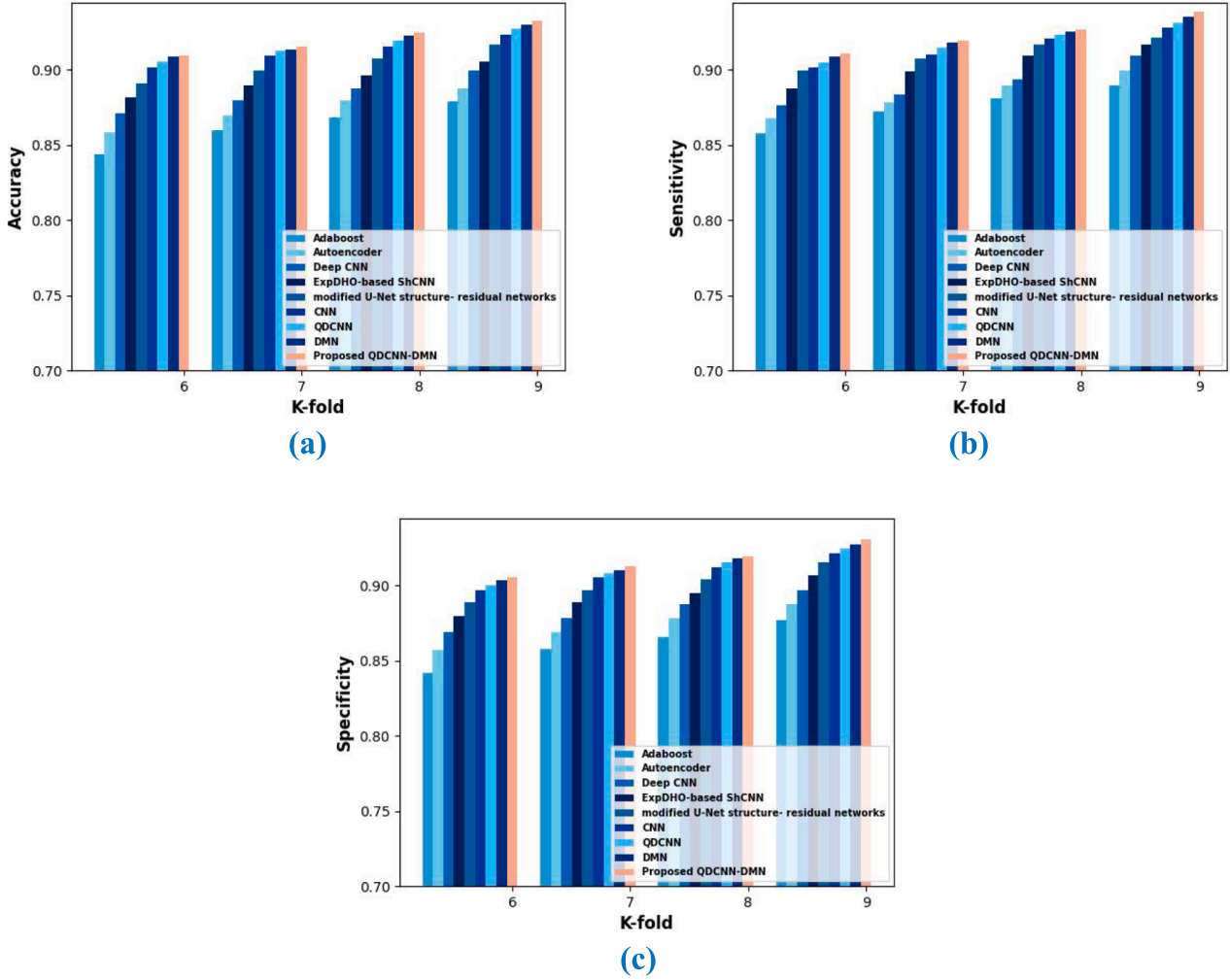


Fig. 12. Comparative evaluation of QDCNN-DMN based on Figshare using k-fold, a) specificity, b) sensitivity and c) accuracy.

QDCNN-DMN layer output, which contains weighted features at different time intervals is represented using the subsequent equations.

The outcome of the QDCNN-DMN layer at the t^{th} interval is expressed by,

$$\gamma = \sum_{c=1}^3 \sum_{y=1}^{q'} \sum_{l=1}^{l'} w'_{c'yl} \mathfrak{V}_{l'c'yl} \quad (23)$$

Here, weights are represented w' and the texture feature is denoted as $\mathfrak{V}_{l'}$. Moreover, the total number of channels is signified as c' and size of DBRIEF, and SLIF are indicated as q' and l' . At the time $(t-1)$, the output is obtained by contemplating the statistical features. Thus, statistical features are considered in which the value of c' changes from 1 to 5. Additionally, the output of the QDCNN-DMN layer at the $(t-1)^{th}$ interval is stated as,

$$\gamma_1 = \sum_{c'=1}^5 K_{c'} * w'_{c'} \quad (24)$$

FC is a branch of applied mathematics that seeks to solve integral and derived expressions by using the Laplace transform. Laplace domain is used to solve the unknown function, and the inverse is then used to get the real data. Here, FC is used to combine the outputs at various instances and from FC [46],

$$L'(t+1) = M'L'(t) + \frac{1}{2}M'L'(t-1) + \frac{1}{6}(1-M')L'(t-2) \quad (25)$$

Employing the consistent outputs of the QDCNN-DMN layer, the above-mentioned expression becomes,

$$\mathfrak{R}_2 = M'\gamma + \frac{1}{2}M'\gamma_1 + \frac{1}{6}(1-M')\mathfrak{R}_1 \quad (26)$$

Substituting the values of γ , γ_1 ,

$$\mathfrak{R}_2 = M' \sum_{c=1}^2 \sum_{y=1}^{q'} \sum_{l=1}^{l'} w'_{c'yl} \mathbb{F}_{c'yl} + \frac{1}{2}M' \times \sum_{c'=1}^5 K_{c'} * w'_{c'} + \frac{1}{6}(1-M')\mathfrak{R}_1 \quad (27)$$

wherein, the constant is denoted as M' and weight is signified as w' .

3.6.3. DMN model

DMN [41] is a feed-forward structure and it comprises numerous layers that create hidden activations through the Maxout function. DMN is the ability to extract intrinsic features from data. Their architecture enables them to detect intricate patterns and representations, which play a critical role in identifying subtle tumor characteristics in medical images. The complexity of DMN can be controlled through hyper-parameters to achieve the balance between computational efficiency and accuracy. Maxout is used to learn multiple features of the input,

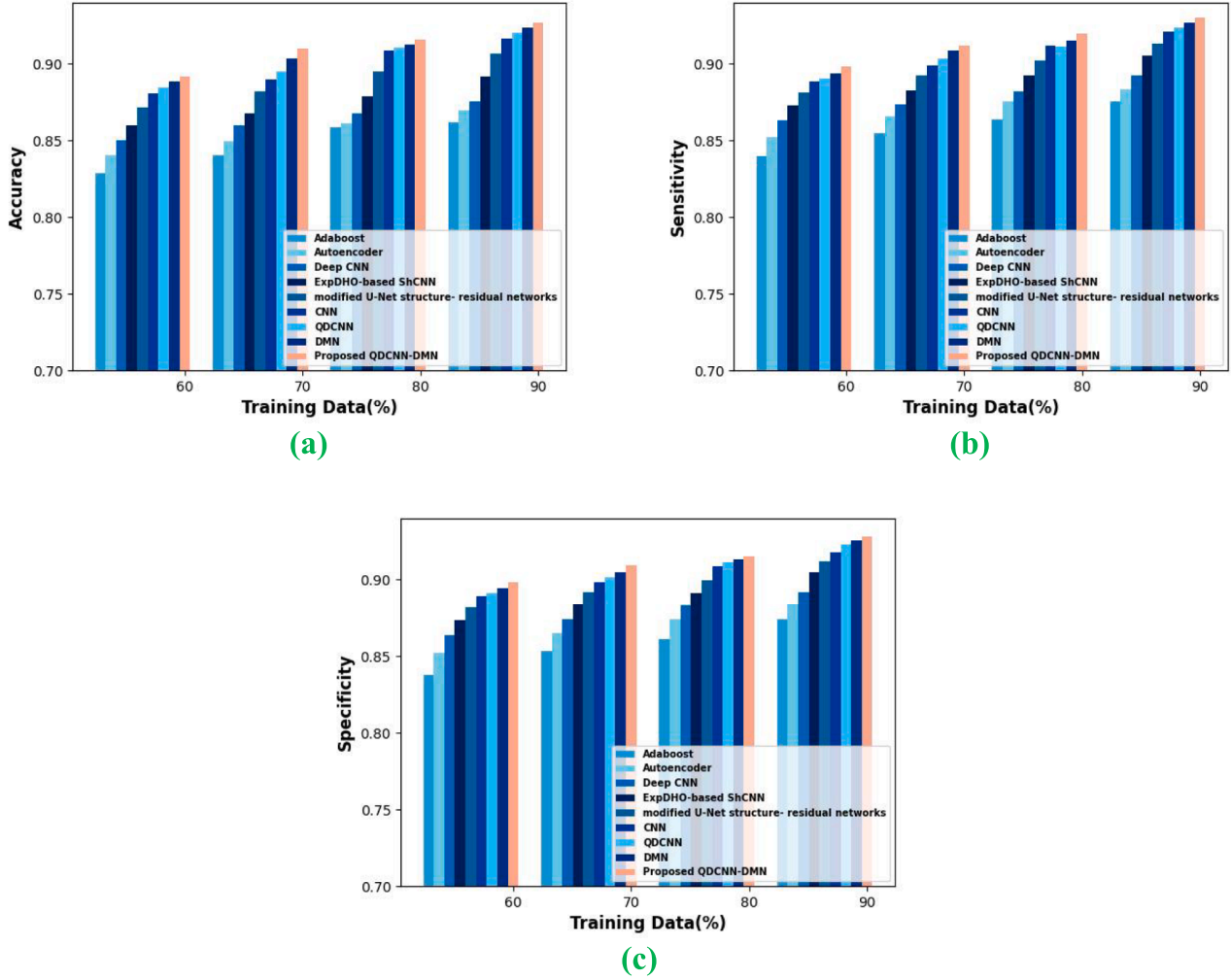


Fig. 13. Comparative evaluation of QDCNN-DMN based on Brain Tumor Classification (MRI) dataset using training data, a) specificity, b) sensitivity and c) accuracy.

enhancing overall efficiency. The input subjected to the DMN technique is signified by X_Z

$$X_Z = \{f_Z, d_Z\} \quad (28)$$

Initially, ReLU is utilized in RBM, as demonstrated as,

$$v_{\mathfrak{Z}} = \begin{cases} \Lambda_{\mathfrak{Z}}, & \text{if } \Lambda_{\mathfrak{Z}} \geq 0 \\ 0, & \text{if } \Lambda_{\mathfrak{Z}} < 0 \end{cases} \quad (29)$$

The above equation $\Lambda_{\mathfrak{Z}}$ designates input given to neurons while $v_{\mathfrak{Z}}$ representing output.

Maxout is a general method of ReLU that becomes the max function on $M(M=2)$ trainable linear operations. For contemplated input $\Lambda \in Z^H$, an output of the Maxout unit is articulated by,

$$v_{\mathfrak{Z}}(\Lambda) = \max_{\psi \in [1, M]} l_{\mathfrak{Z}\psi} \quad (30)$$

Here, $l_{\mathfrak{Z}\psi} = \Lambda^T O_{\dots \mathfrak{Z}\psi} + w_{\mathfrak{Z}\psi}$, $O \in Z^{H \times \Psi \times M}$ and $w \in Z^{\Phi \times \Lambda}$ exposes trainable parameters and count of linear sub-hidden unit designates M .

An activation operation with several structures layer is called DMN. For specified input $\Lambda \in Z^H$, a hidden activation unit is calculated as,

$$A_{\mathfrak{Z}, \psi}^1 = \max_{\psi \in [1, M_1]} \Lambda^T O_{\dots \mathfrak{Z}\psi} + w_{\mathfrak{Z}\psi} \quad (31)$$

$$A_{\mathfrak{Z}, \psi}^2 = \max_{\psi \in [1, M_2]} A_{\mathfrak{Z}, \psi}^1 \Xi O_{\dots \mathfrak{Z}\psi} + w_{\mathfrak{Z}\psi} \quad (32)$$

$$A_{\mathfrak{Z}, \psi}^G = \max_{\psi \in [1, M_G]} A_{\mathfrak{Z}, \psi}^{G-1} \Xi O_{\dots \mathfrak{Z}\psi} + w_{\mathfrak{Z}\psi} \quad (33)$$

$$A_{\mathfrak{Z}, \psi}^{\Psi} = \max_{\psi \in [1, M_{\Psi}]} A_{\mathfrak{Z}, \psi}^{\Psi-1} \Xi O_{\dots \mathfrak{Z}\psi} + w_{\mathfrak{Z}\psi} \quad (34)$$

$$\mathfrak{R}_{\mathfrak{Z}} = \max_{\psi \in [1, M_{\Psi}]} A_{\mathfrak{Z}, \psi}^{\Psi} \quad (35)$$

Here, M_G signifies the number of units in G^{th} the layer while Ψ indicating the total amount of layers in DMN. The expected output achieved from the DMN framework is signified $\mathfrak{R}_{\mathfrak{Z}}$ and the DMN model is illustrated in Fig. 5.

3.7. Classification of Brain tumor utilizing hybrid deep learning

A BT develops when abnormal brain cells proliferate. Tumors can be categorized as either benign or malignant. Malignant BTs originate in the brain and produce faster and violently attack neighboring tissues. It has the potential to spread to other brain regions and impact the central nervous system. The benign BT is a slow-growing mass of cells in the brain. In this work, the tumor is categorized into three types GD- ET, ED, and NCR/NET employing the same QDCNN-DMN, once the tumor is

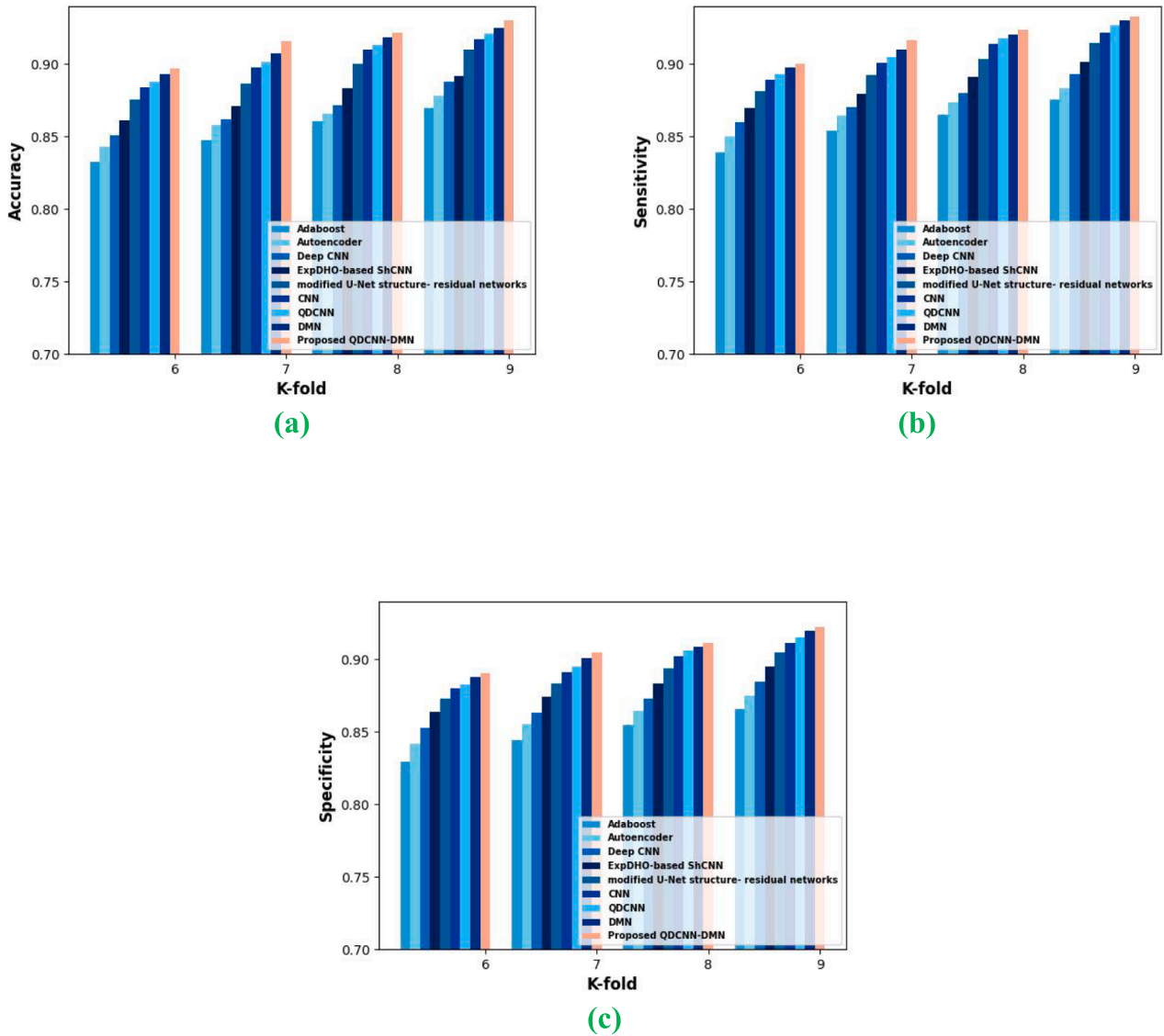


Fig. 14. Comparative evaluation of QDCNN-DMN based on Brain Tumor Classification (MRI) dataset using k-fold, a) specificity, b) sensitivity and c) accuracy.

detected. The QDCNN-DMN is detailed in section 3.7, and the input given for classification is the tumor-detected output ω_C .

4. Results and discussion

The efficiency of QDCNN-DMN for BT classification results is reviewed in this section. Comparison assessment is also investigated and the efficacy of QDCNN-DMN is assessed using existing techniques.

4.1. Experimental setup

Python tool is employed to realize the hybrid QDCNN-DMN for BT classification. The details of the parameters of the developed model are described in Table 2.

4.2. Dataset description

The MRI image used for the implementation was obtained from the BRATS 2018 database and Figshare database and the explanation of the datasets is shown below in the following manner,

a) BRATS 2018 dataset: This dataset [43] uses pre-operative MRI scan images from multiple institutions for segmenting heterogeneous BT-like gliomas. It primarily focuses on predicting the overall survival rate of patients by analyzing a combination of radiomic features. This dataset typically includes 1024 MRI image samples.

b) Figshare dataset: This dataset [42] contains 3064 T1-weighted contrast-enhanced images that were obtained from 233 patients who had three various types of brain malignancies, including gliomas, meningiomas, and pituitary cancers. The average number of imaging slices for meningioma, glioma, and pituitary cancer is 708, 1426 and 930.

c) Brain Tumor Classification (MRI) dataset: In this dataset [47], Brain tumors account for 85 to 90 percent of all primary CNS tumors, 34 percent for men and 36 percent for women. BT is recognized as a highly aggressive diseases affecting both children and adults.

4.3. Evaluation metrics

Performance measures like accuracy, sensitivity, and specificity can be used to evaluate the developed approach.

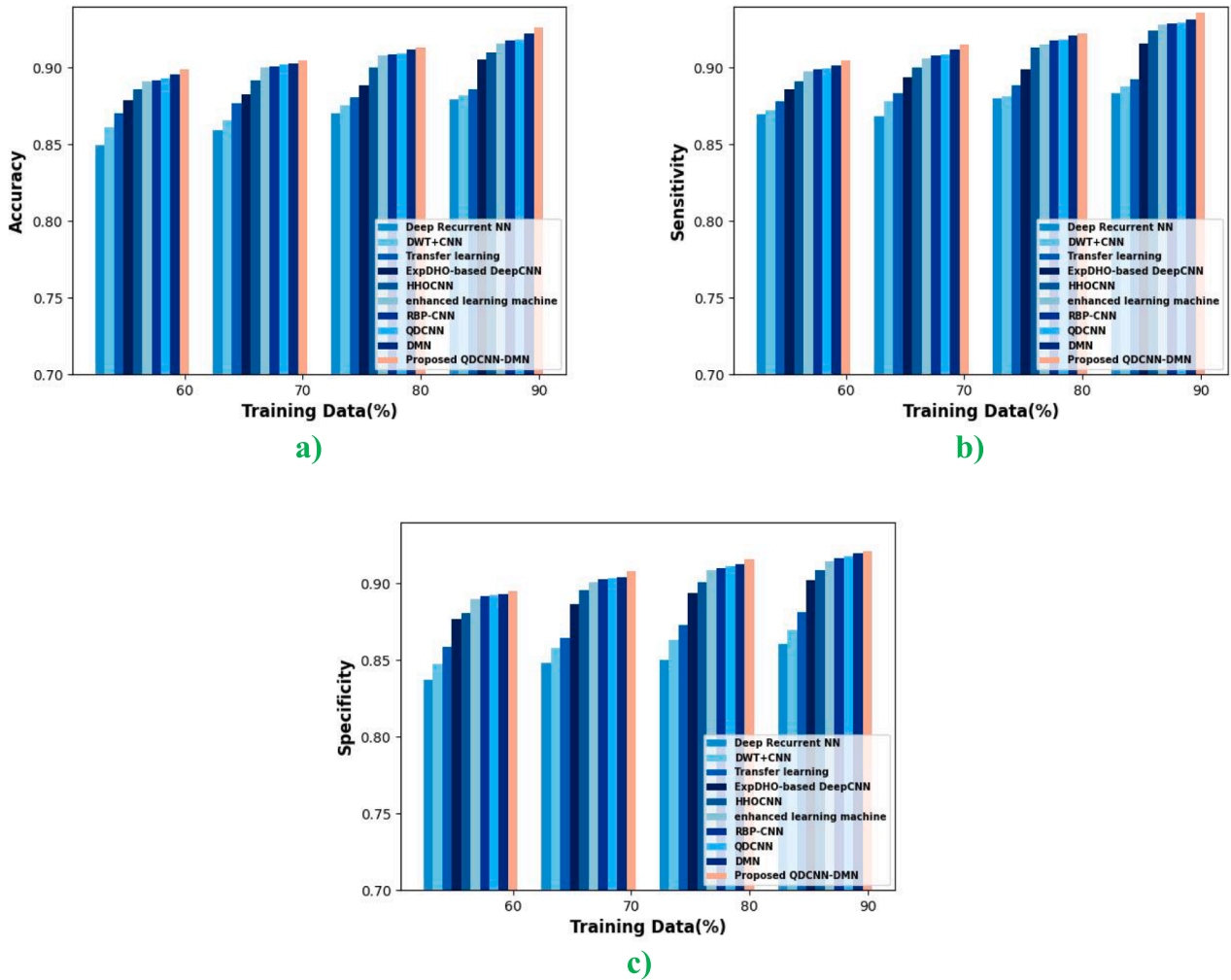


Fig. 15. Investigation of QDCNN-DMN based on BRATS 2018 using training data, a), accuracy, b) sensitivity, and c) specificity.

4.4. Experimental results

The experimental outcomes for the QDCNN-DMN model's classification of BT are briefly elucidated here.

4.4.1. Experimental results with BRATS 2018

An experimental sample image for BT classification utilizing the QDCNN-DMN database from BRATS 2018 is presented in Fig. 6. The input MRI image is demonstrated in Fig. 6a), the enhanced image is displayed in Fig. 6b), the skull stripping image is presented in Fig. 6c), Fig. 6d) shows the segmented image and the image for SLIF feature is depicted in Fig. 6e).

4.4.2. Experimental results with Figshare

The results for classifying BT by applying the Figshare database from QDCNN-DMN are demonstrated in Fig. 7. Fig. 7a), 7b) and 7c) depict input MRI images, enhanced images, segmented images, and skull stripping images respectively. Moreover, segmented image and images for the SLIF feature is displayed in Fig. 7d) and 7e).

4.4.3. Experimental results with Brain tumor classification (MRI) dataset

The experimental results for the BT detection using the Brain Tumor Classification (MRI) dataset from the proposed QDCNN-DMN are illustrated in Fig. 8. Fig. 8a) represents the input of the MRI image, Fig. 8b)

depicts the image of log transform, Fig. 8c) describes the segmented image, Fig. 8d) shows the image of skull stripping, and then Fig. 8e) displays the image for the SLIF feature.

4.5. Comparative methods

The methods taken for evaluating the BT detection process are Autoencoder [26], Adaboost [25], ExpDHO-based ShCNN, Deep CNN [27], modified U-Net structure- residual networks [31], CNN [33], QDCNN [40], and DMN [41] and proposed QDCNN-DMN. Likewise, the numerous comparative methods taken for investigating the BT classification are Deep Recurrent NN [48], DWT + CNN [49], Transfer learning [50], ExpDHO-based Deep CNN, enhanced learning machine [32], HHOCNN [30], QDCNN [40], DMN [41], RBP-CNN [51] and proposed QDCNN-DMN.

4.6. Comparative assessment based on BT detection

The QDCNN-DMN model is investigated for BT detection concerning performance metrics for two different databases such as BRATS 2018, Figshare, and the Brain Tumor Classification (MRI) dataset.

4.6.1. Based on BRATS 2018

(i) Evaluation by varying training data

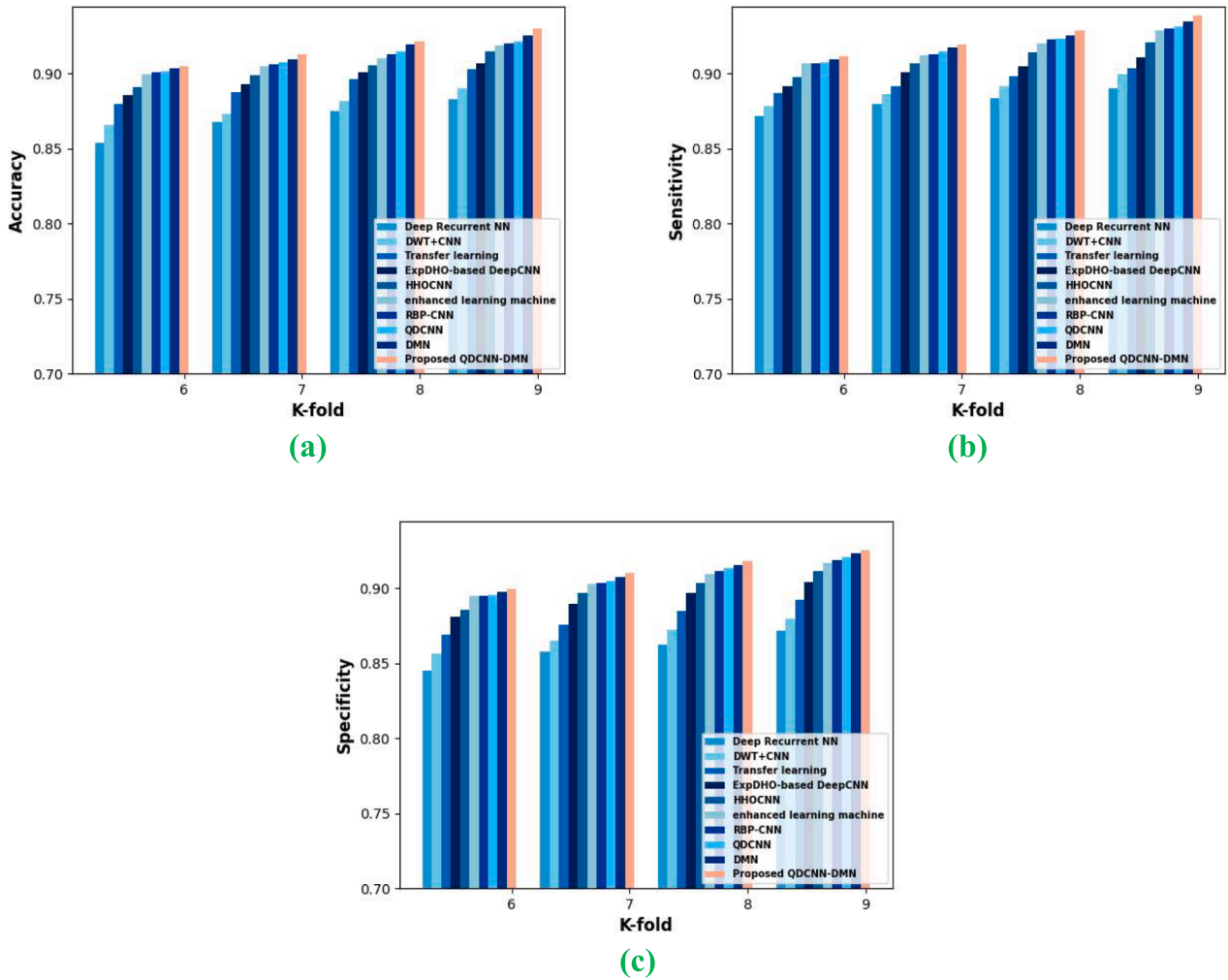


Fig. 16. Investigation of QDCNN-DMN based on BRATS 2018 using k-fold, a), accuracy, b) sensitivity, c) and specificity.

The proposed QDCNN-DMN valuation of the BRATS 2018 database using assessment measures is illustrated in Fig. 9. The accuracy assessment of QDCNN-DMN is demonstrated in Fig. 9a). Accuracy attained by QDCNN-DMN is 92.3%, while the accuracy attained by the prevailing techniques include Adaboost, Autoencoder, Deep CNN, ExpDHO-based ShCNN, modified U-Net structure- residual networks, CNN, QDCNN, and DMN is 86.2%, 86.5%, 88.1%, 90.4%, 91.0%, 91.5%, and 91.9%. Hence, QDCNN-DMN is found to generate a performance enhancement of 6.61%, 6.28%, 4.55%, 2.06%, 1.41%, 0.87%, 0.87%, and 0.43%. Sensitivity assessment of QDCNN-DMN is demonstrated in Fig. 9b). Sensitivity figured by Adaboost, Autoencoder, Deep CNN, ExpDHO-based ShCNN, modified U-Net structure- residual networks, CNN, QDCNN, DMN and QDCNN-DMN is 86.8%, 88.8%, 88.8%, 90.9%, 91.3%, 91.8%, 92.0%, 92.4% and 92.8%, respectively, when considering training data is 90%. This demonstrates that QDCNN-DMN got a performance improvement of 6.47%, 5.17%, 4.31%, 2.05%, 1.62%, 1.08%, 0.86%, and 0.43% respectively. Fig. 9c) demonstrates the valuation of the specificity of QDCNN-DMN. Here, specificity values gained for QDCNN-DMN is 91.8%, whereas traditional approaches gained the values of specificity are 85.2% for Adaboost, 86.0% for Autoencoder, 87.4% for Deep CNN, 89.5% for ExpDHO-based ShCNN, 90.2% for modified U-Net structure- residual networks, 90.8% for CNN, 91.0% for QDCNN, and 91.4% for DMN by considering training data is 90%. This displays that the QDCNN-DMN

reached a performance enhancement of 7.19%, 6.32%, 4.79%, 2.51%, 1.74%, 1.09%, 0.87%, and 0.44% correspondingly.

(ii) Evaluation by varying k-fold

Fig. 10 displays the comparison by varying k-fold based on the BRATS 2018 dataset. Fig. 10 a) demonstrates the accuracy. When the k-fold is 9, then the accuracy of the Adaboost is 87.2%, Autoencoder is 88.8%, Deep CNN is 89.4%, ExpDHO-based ShCNN is 90.2%, modified U-Net structure- residual networks is 90.9%, CNN is 91.8%, QDCNN is 92.1%, and DMN is 92.5% then the accuracy of QDCNN-DMN is 92.7%. The sensitivity graph is given in Fig. 10 b). Here, the sensitivity of the QDCNN-DMN is 93.2%, whereas the sensitivity of the Adaboost, Autoencoder, Deep CNN, ExpDHO-based ShCNN, modified U-Net structure- residual networks, CNN, QDCNN, and DMN is 88.0%, 88.8%, 90.1%, 90.9%, 91.7%, 92.4%, 92.8%, and 93.0% for the k-fold is 9. The specificity is given in Fig. 10 c). The specificity of the Adaboost, Autoencoder, Deep CNN, ExpDHO-based ShCNN, modified U-Net structure- residual networks, CNN, QDCNN, and DMN and the proposed method are 86.2%, 87.4%, 88.4%, 89.9%, 90.6%, 91.3%, 92.2%, 91.6%, and 92.1% for 9 as the k-fold.

4.6.2. Based on the Figshare dataset

(i) Analysis by varying training data

Fig. 11 demonstrates a comparative investigation made by QDCNN-DMN with the Figshare database. The valuation of accuracy is provided

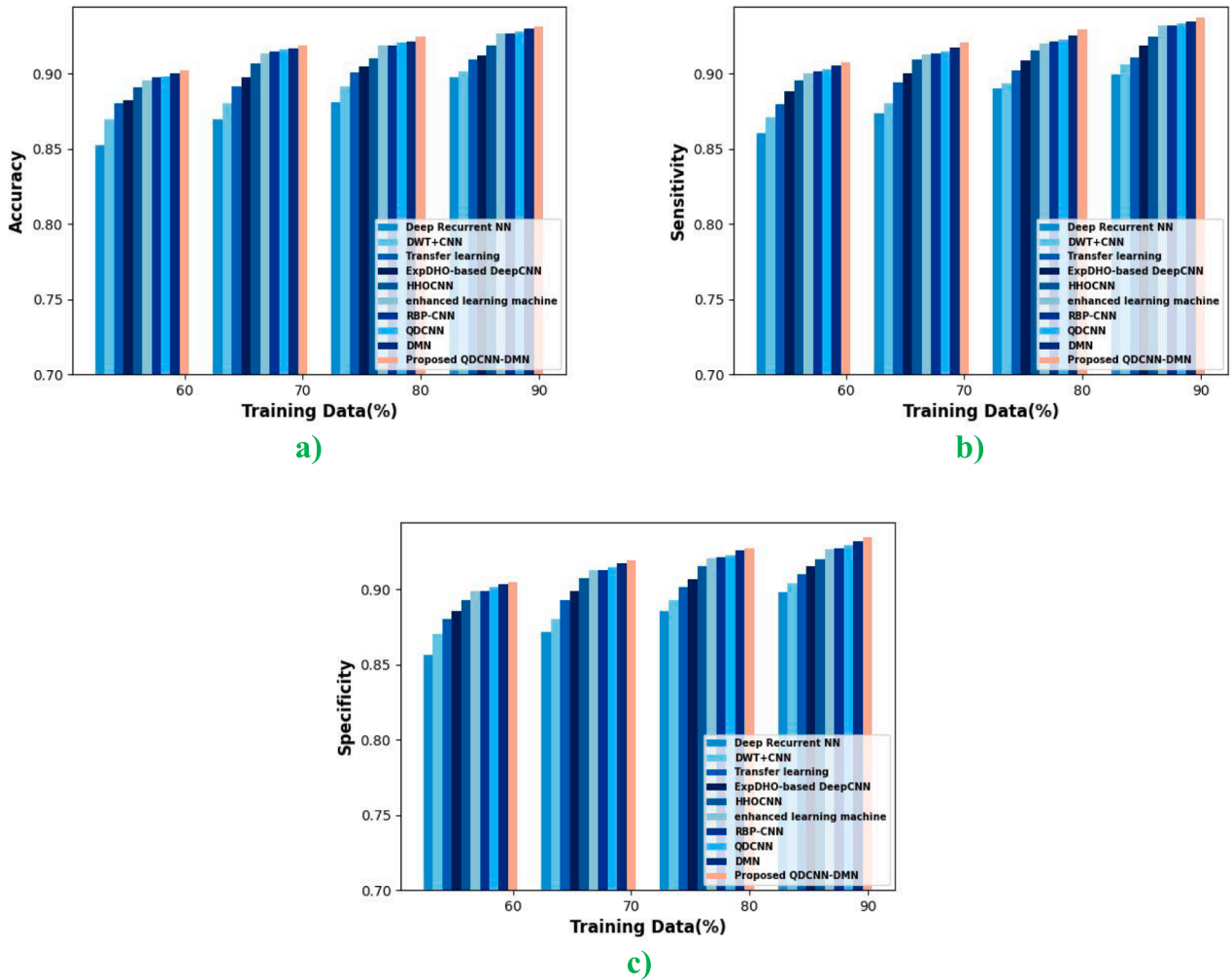


Fig. 17. Investigation of QDCNN-DMN based on Figshare dataset using training data, a) accuracy, b) sensitivity, and c) specificity.

in Fig. 11 a). The training data as 90 %, values of accuracy attained by Adaboost, Autoencoder, Deep CNN, ExpDHO-based ShCNN, modified U-Net structure- residual networks, CNN, QDCNN, DMN and proposed QDCNN-DMN are 88.4 %, 89.5 %, 90.1 %, 90.8 %, 91.5 %, 91.9 %, 92.0 %, 92.5 % and 93.0 %, correspondingly. Furthermore, QDCNN-DMN achieved 4.95 %, 3.76 %, 3.12 %, 2.37 %, 1.61 %, 1.18 %, 1.08 %, and 0.54 % of performance enhancement than conventional methods. The valuation utilizing sensitivity measure is displayed in Fig. 11 b). By considering 90 % of training data, sensitivity gained by Adaboost, Autoencoder, Deep CNN, ExpDHO-based ShCNN, modified U-Net structure- residual networks, CNN, QDCNN, DMN and proposed QDCNN-DMN is 89.0 %, 89.1 %, 90.8 %, 91.6 %, 92.0 %, 92.6 %, 92.9 %, 93.2 % and 93.5 %. The investigation with specificity is demonstrated in Fig. 11 c). When 90 % of training data are considered, the specificity of Adaboost, Autoencoder, Deep CNN, ExpDHO-based ShCNN, modified U-Net structure- residual networks, CNN, QDCNN, DMN and QDCNN-DMN is 87.6 %, 88.8 %, 89.4 %, 90.8 %, 91.4 %, 91.9 %, 92.1 %, 92.4 %, and 92.9 %. In this case, QDCNN-DMN achieved 5.71 %, 4.41 %, 3.77 %, 2.26 %, 1.61 %, 1.08 %, 0.86 %, and 0.54 % of performance enhancement than existing approaches.

(ii) Evaluation by varying k-fold

The assessment by varying k-fold based on the Figshare dataset is explained in Fig. 12. Fig. 12 a) demonstrates the accuracy. The accuracy values of existing methods are 87.9 %, 88.8 %, 90.0 %, 90.6 %, 91.7 %, 92.4 %, 92.8 %, and 93.0 %, then the accuracy of the proposed model is 93.3 %, for the k-fold is 9. Fig. 12 b) depicts the graph between k-fold and sensitivity. Here, the sensitivity of the Adaboost, Autoencoder, Deep CNN, ExpDHO-based ShCNN, modified U-Net structure- residual networks, CNN, QDCNN, DMN and the QDCNN-DMN are 89.0 %, 90.0 %, 91.0 %, 91.7 %, 92.2 %, 93.1 %, 93.1 %, 93.6 %, and 93.9 %, for the k-fold is 9. The specificity of the QDCNN-DMN is given in Fig. 12 c). The specificity values acquired by the Adaboost is 87.7 %, Autoencoder is 88.8 %, Deep CNN is 89.7 %, ExpDHO-based ShCNN is 90.7 %, modified U-Net structure- residual networks is 91.5 %, CNN is 92.1 %, QDCNN is 93.1 %, and DMN is 92.5 %, and the QDCNN-DMN is 92.8 % for 9 as the k-fold.

92.4 %, 92.8 %, and 93.0 %, then the accuracy of the proposed model is 93.3 %, for the k-fold is 9. Fig. 12 b) depicts the graph between k-fold and sensitivity. Here, the sensitivity of the Adaboost, Autoencoder, Deep CNN, ExpDHO-based ShCNN, modified U-Net structure- residual networks, CNN, QDCNN, DMN and the QDCNN-DMN are 89.0 %, 90.0 %, 91.0 %, 91.7 %, 92.2 %, 93.1 %, 93.1 %, 93.6 %, and 93.9 %, for the k-fold is 9. The specificity of the QDCNN-DMN is given in Fig. 12 c). The specificity values acquired by the Adaboost is 87.7 %, Autoencoder is 88.8 %, Deep CNN is 89.7 %, ExpDHO-based ShCNN is 90.7 %, modified U-Net structure- residual networks is 91.5 %, CNN is 92.1 %, QDCNN is 93.1 %, and DMN is 92.5 %, and the QDCNN-DMN is 92.8 % for 9 as the k-fold.

4.6.3. Based on Brain tumor classification (MRI) dataset

(i) Evaluation by varying training data

The evaluation based on the Brain Tumor Classification database by varying the training data is illustrated in Fig. 13. The accuracy assessment is demonstrated in Fig. 13a). Accuracy attained by QDCNN-DMN is 92.7 %, while the accuracy attained by the prevailing techniques include Adaboost, Autoencoder, Deep CNN, ExpDHO-based ShCNN, modified U-Net structure- residual networks, CNN, QDCNN, and DMN is 86.2 %, 87.0 %, 87.5 %, 89.2 %, 90.7 %, 91.7 %, 92.0 %, and 92.3 %. Sensitivity assessment is demonstrated in Fig. 13b). sensitivity figured by Adaboost, Autoencoder, Deep CNN, ExpDHO-based ShCNN, modified U-Net

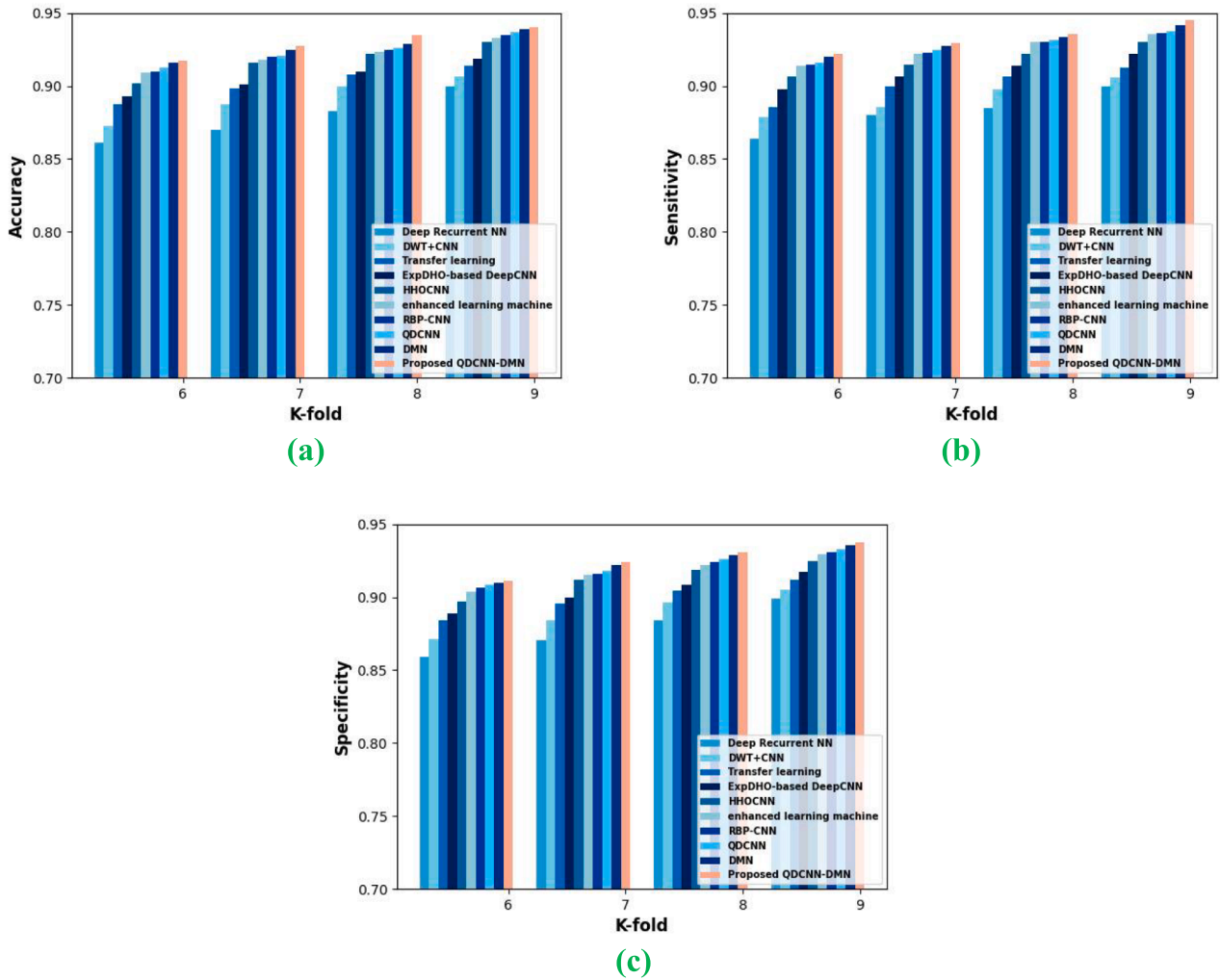


Fig. 18. Investigation of QDCNN-DMN based on Figshare dataset using k-fold, a), accuracy, b) sensitivity, and c) specificity.

structure- residual networks, CNN, QDCNN, DMN and proposed QDCNN-DMN is 87.6 %, 88.3 %, 89.2 %, 90.5 %, 91.3 %, 92.1 %, 92.4 %, 92.7 % and 93.0 % respectively, when considering training data is 90 %. Fig. 13c) demonstrates the valuation of the specificity. Here, specificity values gained for QDCNN-DMN is 92.8 %, whereas the values of specificity are 87.4 % for Adaboost, 88.4 % for Autoencoder, 89.2 % for Deep CNN, 90.5 % for ExpDHO-based ShCNN, 91.2 % for modified U-Net structure- residual networks, 91.8 % for CNN, 92.3 % for QDCNN, and 92.5 % for DMN by considering training data is 90 %.

(ii) Evaluation by varying k-fold

The evaluation of the proposed model based on the Brain Tumor Classification (MRI) dataset by varying the k-fold is shown in Fig. 14. The graph between k-fold and accuracy is shown in Fig. 14 (a). When the k-fold = 9, the QDCNN-DMN measured the value of accuracy is 93.0 % and the Adaboost, Autoencoder, Deep CNN, ExpDHO-based ShCNN, modified U-Net structure- residual networks, CNN, QDCNN, and DMN achieves the accuracy values are 87.0 %, 87.8 %, 88.8 %, 89.2 %, 91.0 %, 91.7 %, 92.1 %, and 92.5 %, respectively. Fig. 14 (b) illustrates the analysis of k-fold with sensitivity. For the value of k-fold = 9, the existing methods measured the sensitivity values are, 87.6 %, 88.3 %, 89.3 %, 90.1 %, 91.4 %, 92.2 %, 92.7 %, and 93.0 %, and then the sensitivity obtained by the QDCNN-DMN is 93.3 %. The graph among the k-value and specificity is described in Fig. 14 (c). Consider the k value is 9, the specificity values of Adaboost is 86.6 %, Autoencoder is

87.5 %, Deep CNN is 88.4 %, ExpDHO-based ShCNN is 89.5 %, modified U-Net structure- residual networks is 90.5 %, CNN is 91.1 %, QDCNN is 91.5 %, DMN is 92.0 %, and QDCNN-DMN is 92.2 %.

4.7. Comparative analysis based on BT classification

The QDCNN-DMN model is investigated for BT classification concerning performance measures for two different databases including BRATS 2018, Figshare, and the Brain Tumor Classification (MRI) dataset.

4.7.1. Based on BRATS 2018 dataset

(i) Analysis by varying training data

Fig. 15 demonstrates the valuation by varying training data from 60 % to 90 % using the BRATS 2018 dataset. The examination of accuracy is given in Fig. 15 a). The accuracy calculated by QDCNN-DMN is 92.6 %, while the Deep Recurrent NN, DWT + CNN, Transfer learning, ExpDHO-based DeepCNN, HHOCNN, enhanced learning machine, RBP-CNN, QDCNN, and DMN calculated accuracy value of 87.9 %, 88.2 %, 88.6 %, 90.5 %, 91.0 %, 91.6 %, 91.8 %, 91.9 %, and 92.2 % for considering 90 % of training data. The performance improvement estimated by QDCNN-DMN in comparison with conventional approaches is 5.08 %, 4.75 %, 4.32 %, 2.27 %, 1.73 %, 1.08 %, 0.86 %, 0.76 %, and 0.43 %. Fig. 15 b) displays an investigation of QDCNN-DMN concerning

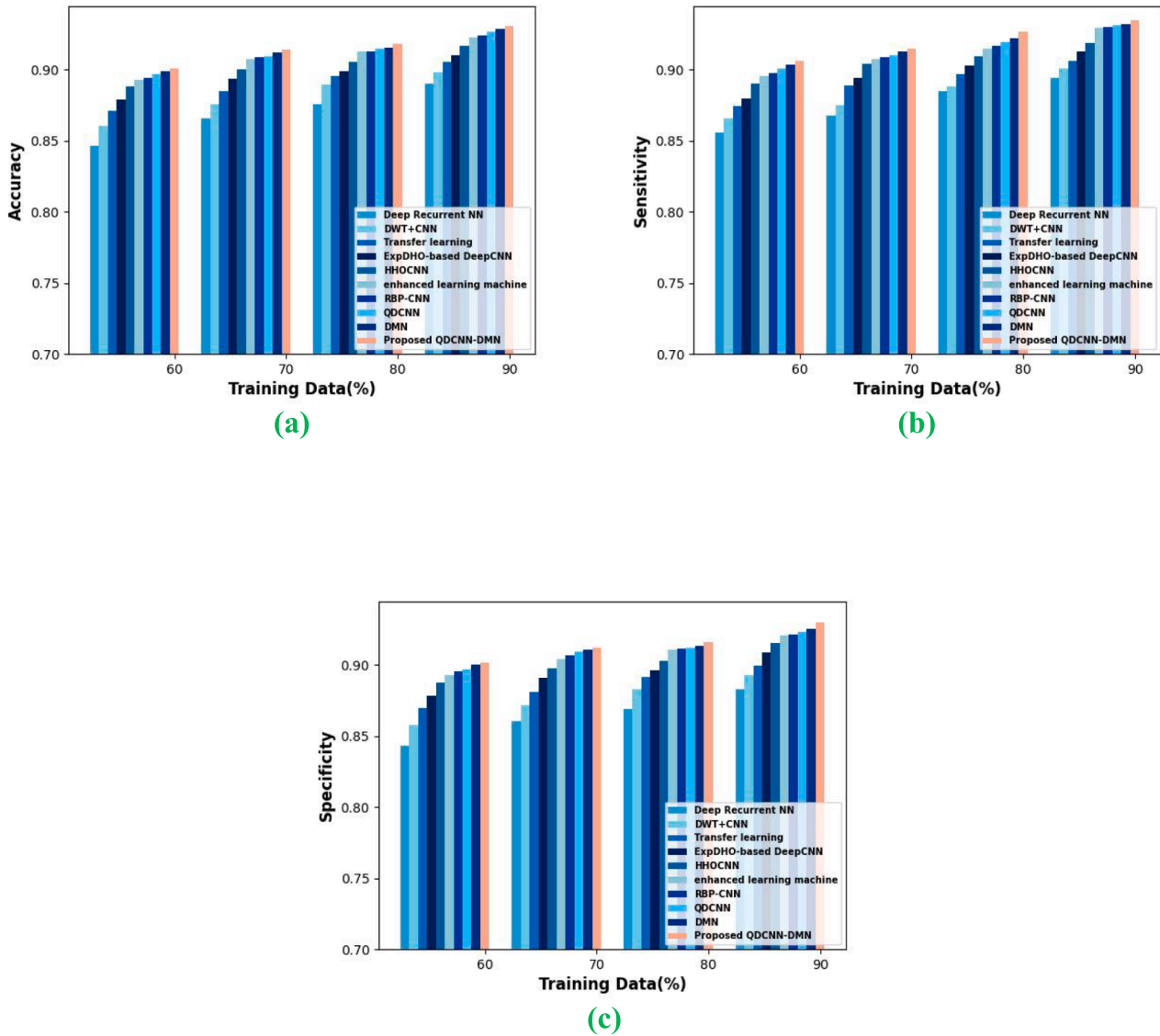


Fig. 19. Investigation of QDCNN-DMN based on Brain Tumor Classification (MRI) dataset using training data, a) accuracy, b) sensitivity, and c) specificity.

sensitivity. For training data is 90 %, the sensitivity value achieved by Deep Recurrent NN is 88.3 %, DWT + CNN is 88.8 %, Transfer learning is 89.3 %, ExpDHO-based DeepCNN is 91.5 %, HHOCNN is 92.4 %, enhanced learning machine is 92.8 %, RBP-CNN is 92.9 %, QDCNN is 92.9 %, DMN is 93.2 % and QDCNN-DMN is 93.5 %. The performance enhancement calculated by the QDCNN-DMN technique when compared with conventional approaches is 5.66 %, 5.13 %, 4.59 %, 2.24 %, 1.18 %, 0.75 %, 0.75 %, 0.64 %, and 0.32 %. Fig. 15 c) gives the valuation of specificity. When assuming training data as 90 %, the specificity value attained by Deep Recurrent NN, DWT + CNN, Transfer learning, ExpDHO-based DeepCNN, HHOCNN, enhanced learning machine, RBP-CNN, QDCNN, DMN, and QDCNN-DMN are 86.0 %, 86.9 %, 88.1 %, 90.2 %, 90.9 %, 91.5 %, 91.7 %, 91.8 %, 92.0 % and 92.1 %. The enhancement of performance attained by QDCNN-DMN when compared with conventional approaches, such as Deep Recurrent NN is 6.62 %, DWT + CNN is 5.65 %, Transfer learning is 4.34 %, and ExpDHO-based DeepCNN is 2.06 %, HHOCNN is 1.3 %, enhanced learning machine is 0.65 %, RBP-CNN is 0.43 %, QDCNN is 0.33 %, and 0.11 % is DMN.

(ii) Evaluation by varying k-fold

Fig. 16 illustrates the comparison of the existing methods with the proposed system by varying k-fold based on the BRATS 2018 dataset. Fig. 16 a) demonstrates the accuracy of the QDCNN-DMN. When the k-fold is 9, then the accuracy of the Deep Recurrent NN is 88.3 %, DWT + CNN is 89.0 %, Transfer learning is 90.3 %, ExpDHO-based DeepCNN is 90.7 %, HHOCNN is 91.5 %, enhanced learning machine is 91.9 %, RBP-CNN is 92.0 %, QDCNN is 92.2 %, and DMN is 92.6 % then the accuracy of QDCNN-DMN is 93.0 %. The sensitivity graph is given in Fig. 16 b). Here, the sensitivity of the QDCNN-DMN is 93.9 %, whereas the sensitivity of the Deep Recurrent NN, DWT + CNN, Transfer learning, ExpDHO-based DeepCNN, HHOCNN, enhanced learning machine, RBP-CNN, QDCNN, and DMN is 89.0 %, 90.0 %, 90.4 %, 91.1 %, 92.1 %, 92.9 %, 93.0 %, 93.2 %, and 93.5 % for the k-fold is 9. The specificity of the QDCNN-DMN is given in Fig. 16 c). The specificity of the Deep Recurrent NN, DWT + CNN, Transfer learning, ExpDHO-based DeepCNN, HHOCNN, enhanced learning machine, RBP-CNN, QDCNN, and DMN and the QDCNN-DMN are 87.2 %, 88.0 %, 89.2 %, 90.4 %, 91.2 %, 91.7 %, 92.5 %, 91.9 %, 92.1 %, and 92.4 % for 9 as the k-fold.

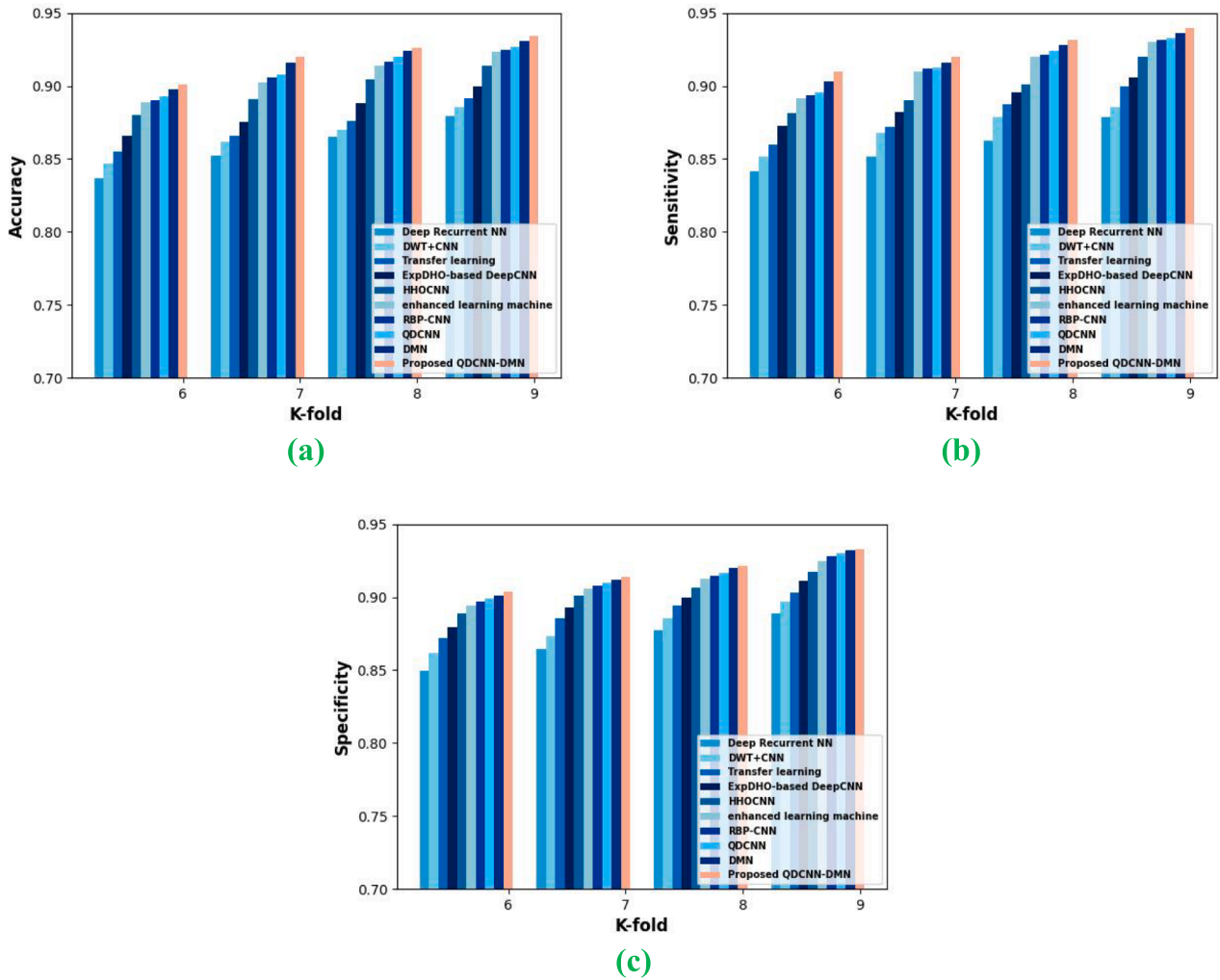


Fig. 20. Investigation of QDCNN-DMN based on Brain Tumor Classification (MRI) dataset using k-fold, a) accuracy, b) sensitivity, and c) specificity.

4.7.2. Based on the Figshare dataset

(i) Assessment by varying training data

Fig. 17 gives a comparative examination of the Figshare database. The valuation of accuracy is given in Fig. 17 a). With training data as 90 %, the accuracy of Deep Recurrent NN, DWT + CNN, Transfer learning, ExpDHO-based DeepCNN, HHOCNN, enhanced learning machine, RBP-CNN, QDCNN, DMN and QDCNN-DMN are 89.8 %, 90.2 %, 91.0 %, 91.2 %, 91.9 %, 92.7 %, 92.7 %, 92.8 %, 93.0 % and 93.2 % correspondingly. Furthermore, QDCNN-DMN achieved 3.65 %, 3.22 %, 2.36 %, 2.15 %, 1.39 %, 0.54 %, 0.54 %, 0.04 %, and 0.21 % of performance enhancement than conventional methods. The assessment in terms of sensitivity measure is displayed in Fig. 17 b). By assuming 90 % of training data, sensitivity gained by Deep Recurrent NN, DWT + CNN, Transfer learning, ExpDHO-based DeepCNN, HHOCNN, enhanced learning machine, RBP-CNN, QDCNN, DMN and QDCNN-DMN is 90.0 %, 90.7 %, 91.1 %, 91.9 %, 92.5 %, 93.2 %, 93.2 %, 93.3 %, 93.5 % and 93.8 %. The performance enhancement of the devised scheme with the existing methods is 4.05 %, 3.30 %, 2.88 %, 2.03 %, 1.39 %, 0.64 %, 0.64 %, 0.53 %, and 0.32 %. The investigation with specificity is demonstrated in Fig. 17 c). For 90 % of training data considered, specificity achieved by Deep Recurrent NN, DWT + CNN, Transfer learning, ExpDHO-based DeepCNN, HHOCNN, enhanced learning machine, RBP-CNN, QDCNN, DMN and QDCNN-DMN is 89.9 %, 90.4 %, 91.0 %, 91.6 %, 92.0 %, 92.7 %, 92.8 %, 92.9 %, 93.2 % and 93.5 %. In this case,

QDCNN-DMN achieved 3.85 %, 3.32 %, 2.67 %, 2.03 %, 1.6 %, 0.86 %, 0.75 %, 0.64 %, and 0.32 % of performance enhancement than existing approaches.

(ii) Evaluation by varying k-fold

The QDCNN-DMN evaluation based on the Figshare dataset by varying the k-fold is shown in Fig. 18. The evaluation of accuracy is explained in Fig. 18 (a). When the k-fold = 9, the QDCNN-DMN achieves an accuracy of 94.0 % and the accuracy of Deep Recurrent NN is 90.0 %, DWT + CNN is 90.7 %, Transfer learning is 91.4 %, ExpDHO-based DeepCNN is 91.9 %, HHOCNN is 93.0 %, enhanced learning machine is 93.3 %, RBP-CNN is 93.5 %, QDCNN is 93.7 %, and DMN is 93.9 %. Fig. 18 (b) represents the graph of sensitivity. The value of k-fold is 9, and the value of sensitivity obtained by the QDCNN-DMN is 94.5 %. Then the existing methods achieve the values of sensitivity are, 90.0 %, 90.6 %, 91.3 %, 92.2 %, 93.0 %, 93.6 %, 93.6 %, 93.8 %, and 94.2 % for Deep Recurrent NN, DWT + CNN, Transfer learning, ExpDHO-based DeepCNN, HHOCNN, enhanced learning machine, RBP-CNN, QDCNN, and DMN. Fig. 18 (c) explains the analysis of k-fold in specificity., the QDCNN-DMN model measured a specificity value is 93.7 %, and then the specificity values of other existing methods are 89.9 %, 90.5 %, 91.2 %, 91.7 %, 92.5 %, 93.0 %, 93.1 %, 93.3 %, and 93.6 % for the k value is 9.

Table 3

Comparative discussion of QDCNN-DMN for BT detection.

Dataset	Metrics	Adaboost	Autoencoder	DeepCNN	ExpDHO-based ShCNN	modified U-Net structure- residual networks	CNN	QDCNN	DMN	Proposed QDCNN-DMN
BRATS 2018	Training Data									
	Accuracy (%)	86.2	86.5	88.1	90.4	91.0	91.5	91.5	91.9	92.3
	Sensitivity (%)	86.8	88.0	88.8	90.9	91.3	91.8	92.0	92.4	92.8
	Specificity (%)	85.2	86.0	87.4	89.5	90.2	90.8	91.0	91.4	91.8
Figshare	Accuracy (%)	88.4	89.5	90.1	90.8	91.5	91.9	92.0	92.5	93.0
	Sensitivity (%)	89.0	89.1	90.8	91.6	92.0	92.6	92.9	93.2	93.5
	Specificity (%)	87.6	88.8	89.4	90.8	91.4	91.9	92.1	92.4	92.9
Brain Tumor Classification (MRI)	Accuracy (%)	86.2	87.0	87.5	89.2	90.7	91.7	92.0	92.3	92.7
	Sensitivity (%)	87.6	88.3	89.2	90.5	91.3	92.1	92.4	92.7	93.0
	Specificity (%)	87.4	88.4	89.2	90.5	91.2	91.8	92.3	92.5	92.8
BRATS 2018	K-fold									
	Accuracy (%)	87.2	88.8	89.4	90.2	90.9	91.8	92.1	92.5	92.7
	Sensitivity (%)	88.0	88.8	90.1	90.9	91.7	92.4	92.8	93.0	93.2
	Specificity (%)	86.2	87.4	88.4	89.9	90.6	91.3	91.6	92.1	92.2
Figshare	Accuracy (%)	87.9	88.8	90.0	90.6	91.7	92.4	92.8	93.0	93.3
	Sensitivity (%)	89.0	90.0	91.0	91.7	92.2	93.1	93.1	93.6	93.9
	Specificity (%)	87.7	88.8	89.7	90.7	91.5	92.1	92.5	92.8	93.1
Brain Tumor Classification (MRI)	Accuracy (%)	87.0	87.8	88.8	89.2	91.0	91.7	92.1	92.5	93.0
	Sensitivity (%)	87.6	88.3	89.3	90.1	91.4	92.2	92.7	93.0	93.3
	Specificity (%)	86.6	87.5	88.4	89.5	90.5	91.1	91.5	92.0	92.2

4.7.3. Based on Brain tumor classification (MRI) dataset

(i) Evaluation by varying training data

Fig. 19 demonstrates the valuation of QDCNN-DMN by varying training data using the Brain Tumor Classification dataset. The examination of accuracy is given in Fig. 19a). The accuracy of QDCNN-DMN is 93.1 %, while Deep Recurrent NN, DWT + CNN, Transfer learning, ExpDHO-based DeepCNN, HHOCNN, enhanced learning, RBP-CNN, QDCNN, and DMN machine calculated accuracy of 89.1 %, 89.9 %, 90.5 %, 91.0 %, 91.7 %, 92.3 %, 92.4 %, 92.7 %, and 92.9 % for considering 90 % of training data. Fig. 19b) displays an investigation of QDCNN-DMN concerning sensitivity. For training data is 90 %, the sensitivity value achieved by Deep Recurrent NN is 89.4 %, DWT + CNN is 90.1 %, Transfer learning is 90.6 %, ExpDHO-based DeepCNN is 91.3 %, HHOCNN is 91.9 %, enhanced learning machine is 92.9 %, RBP-CNN is 93.0 %, QDCNN is 93.2 %, DMN is 93.2 % and QDCNN-DMN is 93.5 %. Fig. 19c) demonstrates the valuation of specificity. When assuming training data as 90 %, specificity value attained by Deep Recurrent NN, DWT + CNN, Transfer learning, ExpDHO-based DeepCNN, HHOCNN, enhanced learning machine, RBP-CNN, QDCNN, DMN and QDCNN-DMN are 88.3 %, 89.3 %, 90.0 %, 90.9 %, 91.5 %, 92.1 %, 92.2 %, 92.4 %, 92.5 % and 93.0 %.

(ii) Evaluation by varying k-fold

The assessment based on the Brain Tumor Classification dataset by varying the k-fold is shown in Fig. 20. Fig. 20 a) describes the analysis of accuracy. The QDCNN-DMN measured an accuracy of 93.4 % and the other existing methods obtained accuracy values are 87.9 %, 88.5 %, 89.2 %, 89.9 %, 91.4 %, 92.3 %, 92.4 %, 92.7 %, and 93.1 % respectively for k-fold = 9. The analysis of sensitivity is explained in Fig. 20 b). When the k-fold is 9, the sensitivity of QDCNN-DMN is 94.0 %. Then the sensitivity is 87.9 %, 88.6 %, 90.0 %, 90.6 %, 92.0 %, 93.0 %, 93.2 %, 93.3 %, and 93.6 % for Deep Recurrent NN, DWT + CNN, Transfer learning, ExpDHO-based DeepCNN, HHOCNN, enhanced learning machine, RBP-CNN, QDCNN, and DMN methods. Fig. 20 c) depicts the graph of specificity. When the k value is 9, the QDCNN-DMN achieved a specificity value is 93.2 %, and then the traditional approaches achieve the value of specificity are 88.9 % for Deep Recurrent NN, 89.7 % for DWT + CNN, 90.3 % for Transfer learning, 91.1 % for ExpDHO-based DeepCNN, 91.7 % for HHOCNN, 92.5 % for enhanced learning machine, 92.8 % for RBP-CNN, 93.0 % for QDCNN, and 93.0 % for DMN.

Table 4

Comparative discussion of QDCNN-DMN for BT classification.

Dataset	Metrics	Deep Recurrent NN	DWT + CNN	Transfer learning	ExpDHO-based Deep CNN	HHOCNN	Enhanced learning machine	RBP-CNN	QDCNN	DMN	Proposed QDCNN-DMN
BRATS 2018	Training Data										
	Accuracy (%)	87.9	88.2	88.6	90.5	91.0	91.6	91.8	91.9	92.2	92.6
	Sensitivity (%)	88.3	88.8	89.3	91.5	92.4	92.8	92.9	92.9	93.2	93.6
	Specificity (%)	86.0	86.9	88.1	90.2	90.9	91.5	91.7	91.8	92.0	92.1
Figshare	Accuracy (%)	89.8	90.2	91.0	91.2	91.9	92.7	92.7	92.8	93.0	93.2
	Sensitivity (%)	90.0	90.7	91.1	91.9	92.5	93.2	93.2	93.3	93.5	93.8
	Specificity (%)	89.9	90.4	91.0	91.6	92.0	92.7	92.8	92.9	93.2	93.5
Brain Tumor Classification (MRI)	Accuracy (%)	89.1	89.9	90.5	91.0	91.7	92.3	92.4	92.7	92.9	93.1
	Sensitivity (%)	89.4	90.1	90.6	91.3	91.9	92.9	93.0	93.2	93.2	93.5
	Specificity (%)	88.3	89.3	90.0	90.9	91.5	92.1	92.2	92.4	92.5	93.0
BRATS 2018	K-fold										
	Accuracy (%)	88.3	89.0	90.3	90.7	91.5	91.9	92.0	92.2	92.6	93.0
	Sensitivity (%)	89.0	90.0	90.4	91.1	92.1	92.9	93.0	93.2	93.5	93.9
	Specificity (%)	87.2	88.0	89.2	90.4	91.2	91.7	91.9	92.1	92.4	92.5
Figshare	Accuracy (%)	90.0	90.7	91.4	91.9	93.0	93.3	93.5	93.7	93.9	94.0
	Sensitivity (%)	90.0	90.6	91.3	92.2	93.0	93.6	93.6	93.8	94.2	94.5
	Specificity (%)	89.9	90.5	91.2	91.7	92.5	93.0	93.1	93.3	93.6	93.7
Brain Tumor Classification (MRI)	Accuracy (%)	87.9	88.5	89.2	89.9	91.4	92.3	92.4	92.7	93.1	93.4
	Sensitivity (%)	87.9	88.6	90.0	90.6	92.0	93.0	93.2	93.3	93.6	94.0
	Specificity (%)	88.9	89.7	90.3	91.1	91.7	92.5	92.8	93.0	93.2	93.2

4.8. Comparative discussion

Table 3 shows the values achieved by the QDCNN-DMN technique for BT detection and classification when compared with prevailing methods considering performance measures like specificity, sensitivity and accuracy. This comparative discussion shows that with the Figshare dataset, the QDCNN-DMN gained a high specificity of 93.1 %, a sensitivity of 93.9 % and an accuracy of 93.3 % when considering the k-fold is 9.

Table 4 shows that the values attained by the QDCNN-DMN method when compared with conventional methods considering performance measures include specificity, sensitivity and accuracy. This comparative discussion demonstrates that with the Figshare database by varying k-fold is 9, QDCNN-DMN attained a specificity of 93.7 %, sensitivity of 94.5 % and accuracy of 94.0 %. The usage of QDCNN-DMN effectively enabled the identification and classification of the BT by learning representative features in the image. Further, the application of skull stripping and segmentation using DEC and SCAN, respectively assisted in the precise detection of tumors, thereby contributing to improved performance.

[The analysis based on t-SNE plot, accuracy vs loss curves, AUC curve, Confusion matrix, and time complexity is provided in the appendix section.].

5. Conclusion

Identifying BT accurately and quickly is crucial to initiating an efficient course of treatment for this disease. An efficient BT detection and classification method for identifying and classifying BT is presented in this work. The input MRI is used for image enhancement, where it is sharpened using logarithmic transformations. DEC is applied in the skull-stripping process. Thereafter, tumor segmentation is carried out by utilizing SCAN, for medical image segmentation. The feature extraction is performed and then, the BT detection is performed by using the hybrid QDCNN-DMN. If the tumor is detected, it is further classified using QDCNN-DMN. It is proved that QDCNN-DMN attained superior specificity, sensitivity and accuracy with 93.7 %, 94.5 % and 94.0 % respectively. The QDCNN-DMN automatically classifies the BT, which enhances the accuracy and reduces the detection time. Also, it is used in the initial identification of the disease, which reduces mortality rates

and saves lives. The limitations of the proposed method are that the prevalence of brain tumor classes in the datasets is uneven. This imbalance could impact the proposed model's ability to learn minority classes, potentially affecting overall performance. Also, in this research, the tumors can occur anywhere in the brain, and their location is often random, which affects the choice of features for classification. The future study will aim to consider the novel filtration method that incorporates adaptive filtering parameters based on dynamic windowing and recurrent mapping for enhancing the performance of BT detection. Also, the future work aims to conduct experiments on many datasets to enhance the generalizability of the technique, and further, use optimization to enhance performance.

Appendix

Abbreviation	Definition
BT	Brain Tumor
QDCNN	Quantum Dilated Convolutional Neural Network
MRI	Magnetic Resonance Imaging
DEC	Deep Embedded Clustering
DMN	Deep Maxout Network
PET	Positron Emission Tomography
MEG	Magneto Encephalography
CT	Computed Tomography
T1w	T1-weighted MRI
PDw	Proton Density-weighted MRI
ML	Machine Learning
DL	Deep Learning
DNNs	Deep Neural Networks
CNN	Convolutional Neural Networks
DWT	Discrete wavelet transform
BRIEF	Binary Robust Independent Elementary Features
SLIF	Spider Local Image Features
LBP	Local Binary Pattern
HoG	Histogram of Gradients
OHDDN	Optimized Hybrid Deep Neural Network
HHOCNN	Harris Hawks optimized the convolution network
KL	Kullback- Leibler
GAN	General Adversarial Networks
FCN	Fully Convolutional Network
LL	Low-Low
HH	High-High
HL	High Low
LH	Low-High
FC	Fractional Calculus
QDC	Quantum Dilated Convolutional
ReLU	Rectified Linear Unit
TNR	True Negative Rate
TPR	True Positive Rate
ExpDHO	Exponential deer hunting optimization
ShCNN	Shepard convolutional neural network
2D	Two-Dimension
GD-ET	Gadolinium –enhancing tumor
ED	peritumoral edema
NCR/NET	necrotic and non-enhancing tumor core
CNS	Central Nervous System
RBP-CNN	regional binary patterns-Convolutional Neural Network
AUC	Area Under the Curve

CRediT authorship contribution statement

Pendela Kanchanamala: Writing – original draft. **Vijayakumar Kuppasamy:** Visualization. **Gunasekaran Ganesan:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

4.9. Analysis of t-SNE plot

Fig. A1 illustrates the analysis of the t-SNE plot. Here, Fig. A1 a) represents the analysis of t-SNE plot using BRATS 2018 dataset; Fig. A1 b) describes the graph of t-SNE plot using Figshare dataset; and Fig. A1 c) depicts the analysis of t-SNE plot based on the Brain Tumor Classification (MRI) dataset. T-SNE plot is a non-linear technique, which is used for data exploration and visualizing high-dimensional data. It is a crucial tool for understanding the complex data structures and revealing the hidden patterns. A t-SNE plot shows the similarities between data points from the original dataset. It places data points close together in a two- or three-dimensional space, capturing their underlying relationships.

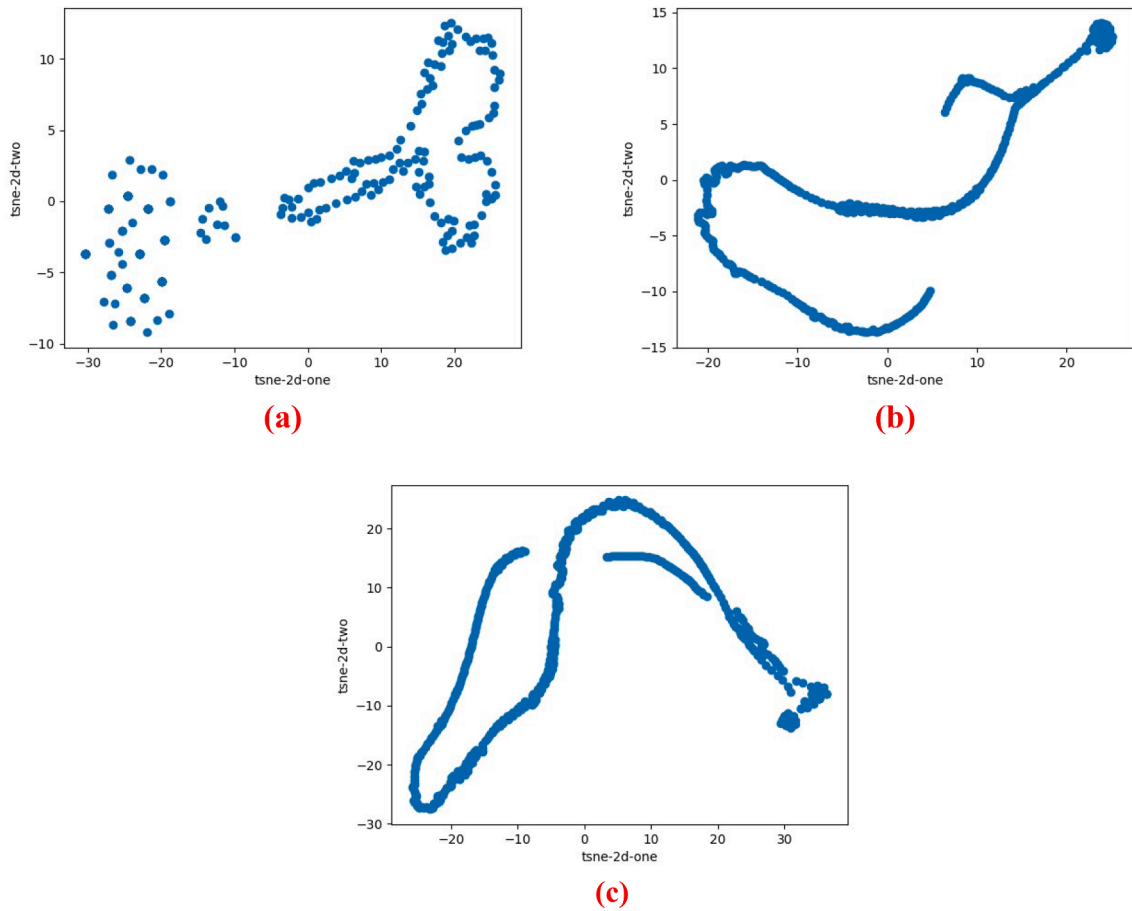


Fig. A1. Analysis of t-SNE using, a) BRATS 2018 dataset, b) Figshare dataset, and c) Brain Tumor Classification (MRI) dataset.

4.10. Analysis of accuracy vs loss curves

Fig. A2 represents the analysis of accuracy vs loss curves. The accuracy vs loss graphs of three datasets like BRATS 2018 dataset, the Figshare dataset, and the Brain Tumor Classification (MRI) dataset is provided in Fig. A2a), A2b), and A2c), respectively. The training performance of the QDCNN-DMN is assessed using the accuracy vs. loss curve. In this case, the loss curve shows the actual difference among the model predicted and actual true output, while the accuracy curve shows how good the model is at predicting the provided data. Here, when the epoch is 5, the value of the QDCNN-DMN is 0.217. Likewise, when the epoch is 10, the value of the QDCNN-DMN is 0.156 based on the BRATS 2018 dataset.

4.11. Analysis of AUC curve

Fig. A3 illustrates the analysis of the AUC curve. The AUC curve based on three datasets like BRATS 2018 dataset, the Figshare dataset, and the Brain Tumor Classification (MRI) dataset is provided in Fig. A3a), A3b), and A3c), respectively. The AUC is a graphical representation that evaluates the performance of a binary classification model across different classification thresholds. It assesses the model's ability to distinguish between two classes: the positive class i.e. indicating the presence of a disease and the negative class i.e. indicating the absence of a disease.

4.12. Confusion matrix

Fig. A4 represents the analysis of confusion matrix. Fig. A4 a) illustrates the graph of confusion matrix using the BRATS 2018 dataset. Fig. A4 b) shows the graphical representation of confusion matrix using Figshare dataset. Then, the Fig. A4 c) presents the analysis of confusion matrix using Brain Tumor Classification (MRI) dataset. A confusion matrix is a tabular summary that showcases the outcomes of prediction and classification tasks. It compares the true labels with the predicted labels generated by a classification model for the target variable. In Fig. A4, the column represents the predicted label of the target variable and the row represents the true label of the target variable.

4.13. Analysis of time complexity

The analysis based on time complexity is shown in Table A1. Here, the time complexity of the previous existing methods such as Deep Recurrent NN, DWT + CNN, Transfer learning, ExpDHO-based Deep CNN, HHOCNN, and Enhanced learning machine are compared with the QDCNN-DMN. Here, when the training data is 90, the existing methods attain the values of Deep Recurrent NN is 44.892sec, DWT + CNN is 41.832sec, Transfer learning is 36.938sec, ExpDHO-based Deep CNN is 34.839sec, HHOCNN is 30.938sec, and Enhanced learning machine is 25.823sec, and then the value of QDCNN-DMN is 18.533sec, based on the BRATS 2018 dataset. The time complexity of the QDCNN-DMN is minimal when compared to all existing models.

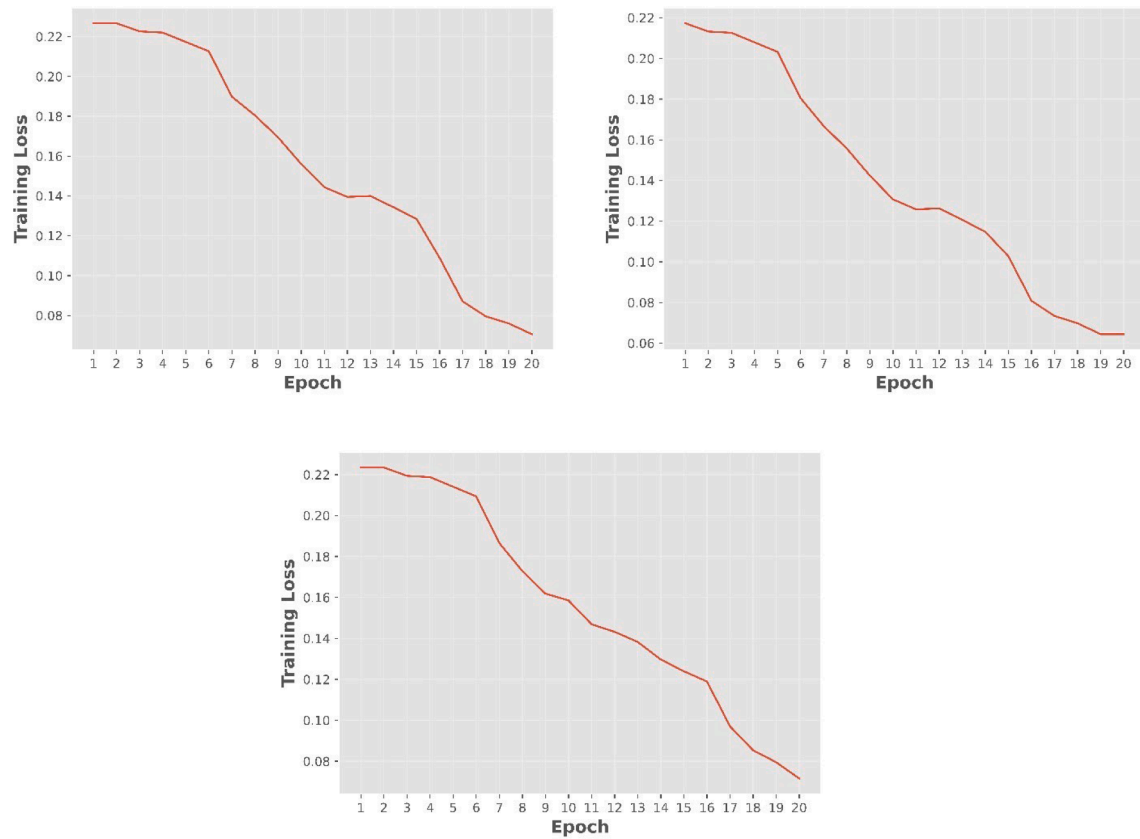


Fig. A2. Analysis of accuracy vs loss curves using, BRATS 2018 dataset, Figshare dataset, and Brain Tumor Classification (MRI) dataset.

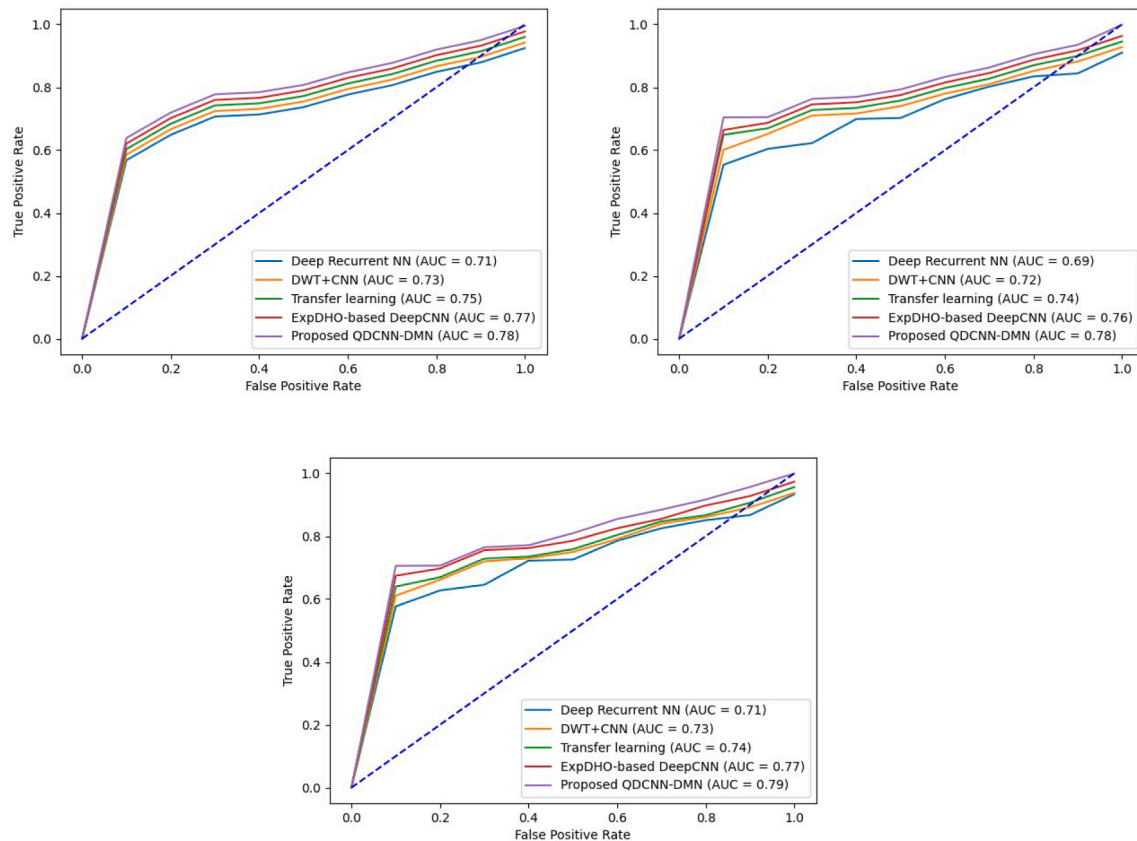


Fig. A3. Analysis of AUC curve using BRATS 2018 dataset, Figshare dataset, and Brain Tumor Classification (MRI) dataset.

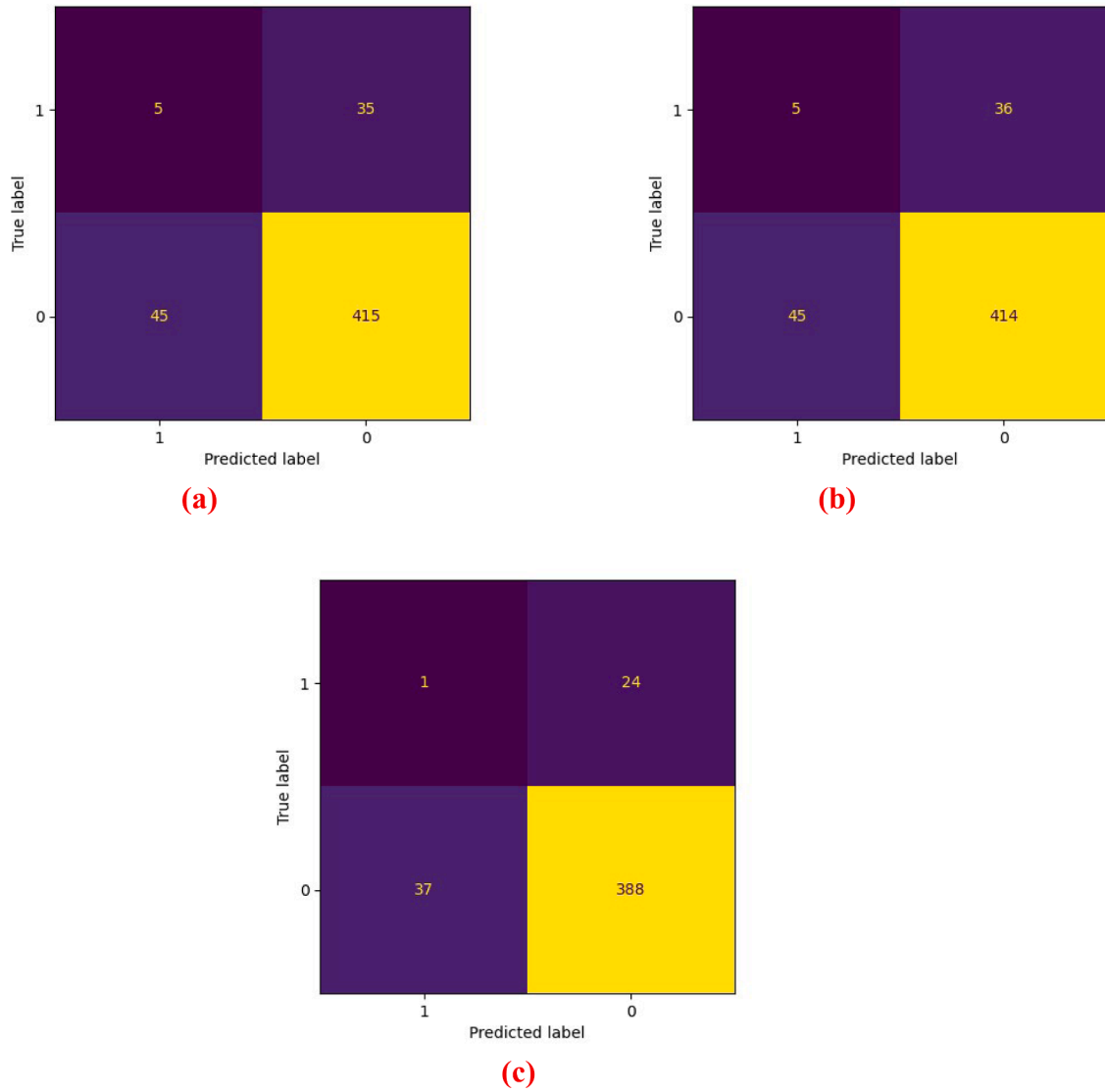


Fig. A4. Analysis of confusion matrix using, a) BRATS 2018 dataset, b) Figshare dataset, and c) Brain Tumor Classification (MRI) dataset.

Table A1

Analysis of time complexity.

Training data/ Methods	Deep Recurrent NN	DWT + CNN	Transfer learning	ExpDHO-based Deep CNN	HHOCNN	Enhanced learning machine	Proposed QDCNN- DMN
BRATS 2018 dataset (sec)							
60	34.829	30.833	26.938	22.839	17.939	13.675	9.214
70	37.839	34.839	30.958	26.949	21.833	16.674	13.644
80	41.829	38.829	32.839	30.847	26.938	20.765	15.744
90	44.892	41.832	36.938	34.839	30.938	25.823	18.533
Figshare dataset (sec)							
60	31.829	27.921	23.949	19.839	14.893	9.839	5.782
70	35.839	31.982	26.954	23.894	18.839	14.933	8.839
80	38.839	34.839	32.464	27.933	22.765	17.839	11.839
90	42.829	38.837	35.567	31.93	25.932	21.899	15.839
Brain Tumor Classification (MRI) dataset (sec)							
60	32.829	28.829	25.4435	21.339	16.416	11.757	7.498
70	35.839	31.829	28.956	25.4215	20.336	15.8035	11.2415
80	37.943	36.839	33.6515	29.39	24.8515	19.302	13.7915
90	42.821	40.983	37.7525	33.3845	28.435	23.861	17.186

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